Étale Fundamental Groups

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Boring admin stuff

We're meeting every Thursday at 13:00-14:00 in J11 Hicks (barring strike days), beginning 8th October. Notes will be updated (hopefully) shortly after each talk, and will be emailed out once they're finished, also available on the GitHub page. The main reference is Section 5 of Tamás Szamuley's *Galois Groups and Fundamental Groups* [Sza09].

Per the schedule, the current talk outline is:

1. Motivation and examples (Mark)
2. Schemes and morphisms (Diego, split into two talks for time constraints)
3. Topological fundamental groups as automorphism groups (Pierre-Louis)
4. Geometric points and the definition of $\pi_1^{\text{ét}}$ (Brais)
5. Étale deck transformations and pro-objects (Matt)
6. Examples (Matt)
7. Galois theory for finite étale covers (Hao)
8. Outer Galois actions and further examples (Mark).

1 Motivation and examples (Mark)

1.1 Fundamental groups and covers: the classical case

Let X be a topological space. One historically important problem is that of classifying the *covering spaces* of X.

Definition 1.1. Suppose X and Y are topological spaces. A continuous map $p : Y \to X$ is called a *cover* if for all $x \in X$, there is an open neighbourhood $U_x \subseteq X$ of x whose preimage decomposes as a disjoint union of homeomorphic copies of U_x, that is, we may decompose

$$p^{-1}[U_x] = \coprod_{i \in I} V_{x,i}$$

with each $V_{x,i} \subseteq Y$ open and $p|_{V_{x,i}} : V_{x,i} \xrightarrow{\sim} U_x$ a homeomorphism. We say that Y is a *covering space* for X. If X and Y are additionally pointed (i.e. have distinguished basepoints $x_0 \in X, y_0 \in Y$), and the cover $p : (Y, y_0) \to (X, x_0)$ satisfies $p(y_0) = x_0$, we may say that (Y, y_0) is a *pointed covering space* of (X, x_0).

These covering spaces of X define a category $\mathbf{Cov}(X)$, whose morphisms are continuous maps $Y_1 \to Y_2$ commuting with the covers $p_i : Y_i \to X$. We will write $\mathbf{FinCov}(X)$ for the full subcategory of finite covers, and $\mathbf{Cov}_*(X, x_0)$ for the subcategory of pointed covers (with pointed morphisms).

One notable property of covering spaces is the *homotopy lifting property*: that given a homotopy $\{f_t : Z \to X\}_{t \in [0,1]}$ of maps to X, and a lift $g_0 : Z \to Y$ of f_0 through $p : Y \to X$, there exists a unique homotopy $\{\hat{f}_t : Z \to Y\}_{t \in [0,1]}$ lifting $\{f_t\}_t$ through $p : Y \to X$ such that $\hat{f}_0 = g_0$.

As a consequence of the homotopy lifting property, their fibres admit a *monodromy action*: given a cover $p : Y \to X$ and some $x_0 \in X$, the fibre $p^{-1}(x_0) \subseteq Y$ is a discrete subspace. Given a point $y \in p^{-1}(x_0)$ and some class $[\alpha] \in \pi_1(X, x_0)$, which we recall is a strong homotopy equivalence class of loops in X based at x_0, any representative $\alpha : [0,1] \to X$ lifts to a unique $\hat{\alpha}_y : [0,1] \to Y$ starting at y whose strong homotopy class depends only on the strong homotopy class $[\alpha]$. Thus the endpoint $\hat{\alpha}_y(1)$ does not depend on the choice of lift, and we define $[\alpha] \cdot y := \hat{\alpha}_y(1)$.

So we can think of the taking the fibre at x_0 as a functor

$$\mathrm{Fib}_{x_0} : \mathbf{Cov}(X) \to \pi_1(X, x_0)\text{-}\mathbf{Set}$$

from the category of covering spaces of X to the category of $\pi_1(X, x_0)$-sets, i.e. discrete sets with an action of $\pi_1(X, x_0)$. As it turns out, when X is especially nice (path-connected, locally path-connected, and semilocally simply connected) this is a *representable functor*, insomuchas

$$\mathrm{Fib}_{x_0}(Y) \cong \mathrm{Hom}_{\mathbf{Cov}(X)}(\tilde{X}, Y)$$

for some $\tilde{X} \in \mathbf{Cov}(X)$ unique up to isomorphism. Such $\tilde{X}$ is called a *universal covering space* for X, and may be constructed explicitly by choosing a basepoint $x_0 \in X$ and defining

$$\tilde{X}_{x_0} := \frac{\{\gamma : [0,1] \xrightarrow{\text{cts}} X \mid \gamma(0) = x_0\}}{\text{homotopy}}, \qquad \tilde{p} : \tilde{X}_{x_0} \to X;\ \gamma \mapsto \gamma(1).$$

By definition, we see that $\mathrm{Fib}_{x_0}(\tilde{X}_{x_0}) = \pi_1(X, x_0)$, and thus by universality we see

$$\pi_1(X, x_0) \cong \mathrm{Hom}_{\mathbf{Cov}(X)}(\tilde{X}_{x_0}, \tilde{X}_{x_0}) = \mathrm{Aut}_{\mathbf{Cov}(X)}(\tilde{X}_{x_0})$$

This final isomorphism is sometimes referred to as the *deck transformation construction*.

Theorem 1.2. *Pointed covers of a path-connected, locally path-connected, and semilocally simply connected pointed space (X, x_0) are in bijection with subgroups of $\pi_1(X, x_0)$. To each $H \leq \pi_1(X, x_0)$ we associate $(\tilde{X}_{x_0}/H, Hx_0)$, and to each $p : (Y, y_0) \to (X, x_0)$ we associate $p_*\pi_1(Y, y_0) \leq \pi_1(X, x_0)$.*

The main idea of this reading group will be to define a sort of covering space theory for schemes, by way of the *étale fundamental group*. But how do we do this?

Topological spaces	Schemes
X a nice topological space	S a nice scheme
$p : Y \to X$ a topological cover	$\phi : X \to S$ a *finite étale cover*
$x_0 \in X$ a point	$\overline{s_0} : \mathrm{Spec}(\Omega) \to S$ a *geometric point*
$\pi_1(X, x_0)$ defined in terms of loops/paths	Paths are meaningless in Zariski!
$\pi_1(X, x_0) = \mathrm{Aut}(\mathrm{Fib}_{x_0})$	$\pi_1^{\text{ét}}(S, \overline{s_0}) = \mathrm{Aut}(\mathrm{Fib}_{\overline{s_0}})$
$\tilde{X} \to X$ universal cover (generally infinite degree)	$\mathfrak{X} \in \mathrm{pro}(\mathbf{Fét}_S)$ a *universal pro-object*
$\pi_1(X, x_0) \cong \mathrm{Aut}(\tilde{X})$	$\pi_1^{\text{ét}}(S, s_0) \cong \varprojlim_{X \in \mathfrak{X}} \mathrm{Aut}(X)$

In fact for complex affine varieties $X_{/\mathbb{C}}$ we will be able able to define both the classical fundamental group $\pi_1(X^{\text{an}}, x_0)$ (considering its complex points $X(\mathbb{C})$ canonically retopologised as a complex manifold X^{an}) and the étale fundamental group $\pi_1^{\text{ét}}(X, \overline{x_0})$ (considering it as a scheme, where $\overline{x_0}$ is the geometric point induced by x_0), and in this case the two *almost* coincide:

$$\pi_1^{\text{ét}}(X, \overline{x_0}) \cong \widehat{\pi_1}(X^{\text{an}}, x_0),$$

where the only difference is a profinite completion on the right, appearing since we are only considering *finite* covers as we will see later.

The approach we will take is to use this recontextualisation of $\pi_1(X, x_0)$ as an automorphism group as the basis for our definition of $\pi_1^{\text{ét}}$. After defining all of the scheme theory we will need, we will try to reformulate the theory of covering spaces independent of loops so we can do it for schemes. There will be some interesting differences which will be explained in future talks.

1.2 An interesting coincidence

One slightly more specific example which might motivate the need for a theory of étale covering space is the famous *Weil Rosetta stone*. This is a set of strange coincidences between the theories of Riemann surfaces and of number fields that seem to suggest that both are offshoots of a more general theory (which will turn out to be our theory of the étale fundamental group). To state this, we first must define some things. We will start with the number fields setting.

1.2.1 A little bit of number theory

Definition 1.3. A *number field* K is a finite field extension of $\mathbb{Q}$. That is, it is a field containing $\mathbb{Q}$ which is finite-dimensional as a $\mathbb{Q}$-vector space. Each number field K admits a distinguished subring $\mathcal{O}_K$ called its *ring of integers*, which is the integral closure of $\mathbb{Z}$ in K.

In each extension L/K of number fields, we can look at prime ideals $\mathfrak{p}$ of the ring of integers $\mathcal{O}_K$ of K (which we be called 'primes of K' for short). Each such prime must factorise in $\mathcal{O}_L$ as a unique product

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}$$

of powers of primes $\mathfrak{P}_i$ of L. We say that $\mathfrak{p}$ *ramifies* in L/K if any of the $e_i > 1$, otherwise we say it is *unramified*.

Example 1.4. In the extension $\mathbb{Q}(i)/\mathbb{Q}$, the ideal $5\mathbb{Z}[i]$ factorises as $\langle 2+i\rangle\langle 2-i\rangle$, so 5 is unramified. In contrast, the ideal $2\mathbb{Z}[i]$ factorises as $\langle 1+i\rangle^2$ since $(1+i)(1-i) = 2$, so 2 ramifies. ♢

If a set S of primes of K contains all of the primes at which L/K ramifies, we say that L/K is *unramified away from S*, or just *S-unramified*.

Given an algebraic closure $\overline{K}/K$, we can form the *maximal S-unramified extension* K_S/K by taking the union of all finite subextensions L/K of $\overline{K}/K$ which are S-unramified. Then we define the *S-unramified absolute Galois group* $G_{K,S}$ to be topological group

$$G_{K,S} := \mathrm{Gal}(K_S/K) := \varprojlim_{L/K} \mathrm{Aut}_K(L)$$

where L/K ranges over all finite, S-unramified extensions.

Theorem 1.5. *The open subgroups of $G_{K,S}$ are in bijection with the set of finite, S-unramified extensions of K contained in $\overline{K}$. An open subgroup H is sent to its fixed field K_S^H, and a finite S-unramified extension L/K is sent to the open subgroup* $\mathrm{Gal}(K_S/L)$.

There is an underlying geometric object here, called $\mathrm{Spec}(\mathcal{O}_{K,S})$ (see next week's talk). We will see that finite S-unramified extensions correspond to finite étale covers $X \twoheadrightarrow \mathrm{Spec}(\mathcal{O}_{K,S})$, and a choice of algebraic closure $\overline{K}$ corresponds to a choice of *geometric basepoint* $\mathrm{Spec}(\overline{K})$.

1.2.2 A little bit of complex geometry

Definition 1.6. A *Riemann surface* is a smooth complex manifold of dimension 1.

We are interested in holomorphic maps between compact Riemann surfaces, so fix a surjective holomorphic map $\phi : Y \to X$. Interestingly, any such map is a finite topological cover away from a finite set of points of X, which are called the *ramification points* of ϕ. If a discrete subset $S \subset X$ contains all of the ramification points of ϕ, we say that ϕ is *unramified away from S*.

Since any finite cover of Riemann surface is itself a Riemann surface, Theorem 1.2 specialises in this case to give a specific correspondence between pointed, S-unramified holomorphic covers of (X, x_0) and finite-index subgroups of $\pi_1(X \setminus S, x_0)$, which is equivalent to the following statement about the profinite completion $\widehat{\pi_1}$ of the fundamental group.

Theorem 1.7. *The open subgroups of $\widehat{\pi_1}(X \setminus S, x_0)$ are in bijection with finite, S-unramified, pointed holomorphic covers of (X, x_0). An open subgroup H is sent to the compactification of the orbit space $\widetilde{X \setminus S}_{x_0}/H$ with basepoint Hx_0, and a finite, S-unramified, pointed holomorphic cover $\phi : (Y, y_0) \to (X, x_0)$ is sent to the open subgroup*

$$\phi_* \widehat{\pi_1}(Y \setminus \phi^{-1}[S], y_0) \subseteq \widehat{\pi_1}(X \setminus S, x_0).$$

There is an underlying algebraic object here, that being the field $K(X)$ of meromorphic functions on X, and the subring $\mathcal{O}_{X,S}$ of meromorphic functions which are holomorphic away from S. Any holomorphic cover $\phi : Y \to X$ realises a field extension $\phi^* : K(X) \hookrightarrow K(Y)$, and it sends $\mathcal{O}_{X,S}$ to $\mathcal{O}_{Y,\phi^{-1}[S]}$.

1.2.3 Weil's Rosetta stone

We can formalise this correspondence later, but for now there's a long list of things that seem to be similar. We will write the objects of interest, as well as what the 'étale equivalent' will be.

Compact Riemann surface X	**Number field** K	*Étale equivalent*
X		$\mathrm{Spec}(\mathcal{O}_K)$
X'		$\mathrm{Spec}(\mathcal{O}_{K,S})$
$\phi : Y \twoheadrightarrow X$ holomorphic	$\iota : K \hookrightarrow L$ finite extension	$\iota^* : \mathrm{Spec}(\mathcal{O}_L) \twoheadrightarrow \mathrm{Spec}(\mathcal{O}_K)$ finite smooth
S ramification points	S ramified primes	S ramification points
ϕ is S-unramified	L/K is S-unramified	ι^* is S-unramified
$\phi\|_{Y'} \in \mathbf{FinCov}(X \setminus S)$		$\iota^*\|_{\mathrm{Spec}(\mathcal{O}_{L,S})} \in \mathbf{Fét}(\mathrm{Spec}(\mathcal{O}_{K,S}))$
$x_0 \in X'$ basepoint	$\overline{K}/K_S/K$ algebraic closure	$\overline{s_0} : \mathrm{Spec}(\overline{K}) \to \mathrm{Spec}(\mathcal{O}_{K,S})$ geom. point
$\widehat{\pi_1}(X', x_0)$	$G_{K,S} := \mathrm{Gal}(K_S/K)$	$\pi_1^{\text{ét}}(\mathrm{Spec}(\mathcal{O}_{K,S}), \overline{s_0})$
$\phi^{-1}(x)$	Primes above $\mathfrak{p} \notin S$	$\mathrm{Spec}(\mathcal{O}_{L,S}) \times_{\mathrm{Spec}(\mathcal{O}_{K,S})} \mathrm{Spec}(\Omega)$
Classification of hol. covers	Galois theory	Still to come...

Here we write $X' := X \setminus S$ and $Y' := Y \setminus \phi^{-1}[S]$ for brevity's sake.

1.2.4 A bonus "rad" fact

Though not entirely in the spirit of finite étale covers, the analogy between compact Riemann surfaces and number fields is even deeper than this. This subsection is only included for fun and is very informal, so feel free to skip it!

One of the greatest achievements of early 20th Century number theory was the development of *class field theory*. The point of class field theory is to classify every so-called abelian number field, i.e. number fields with abelian Galois group over $\mathbb{Q}$. Along with the development of a theory of analysis over $\mathbb{Q}$ (i.e. analysis over the ring $\mathbb{A}_{\mathbb{Q}}$ of adèles of $\mathbb{Q}$), it is known for some fairly mysterious isomorphisms. One of the most notable being:

$$G_{K,S}^{\text{ab}} \cong K^{\times} \backslash \mathbb{A}_{K,\text{fin}}^{\times} / U_S, \qquad U_S := \prod_{v \notin S} U_v.$$

The double quotient group here may be thought of as the Picard group of $\text{Spec}(\mathcal{O}_{K,S})$, i.e. we may think of this as a canonical isomorphism

$$G_{K,S}^{\text{ab}} \cong \text{Pic}(\text{Spec}\,\mathcal{O}_{K,S}).$$

Simultaenously on the other side, if we take the (punctured) compact Riemann surface $X' := X \setminus S$ we may consider its *Jacobian variety*

$$\text{Jac}(X') := H^0(X', \Omega^1_{X'})^* / H_1(X')$$

where a 1-chain γ is seen as a functional on global holomorphic differentials via the usual pairing $(\gamma, \omega) \mapsto \int_\gamma \omega$. Then *geometric class field theory* gives another mysterious canonical isomorphism

$$\pi_1(X', x_0)^{\text{ab}} \otimes \mathbb{R}/\mathbb{Z} \cong \text{Jac}(X').$$

Since $\text{Jac}(X')$ may be seen as the identity component of the Picard variety, Weil's Rosetta stone suggests that these two statements are somewhat analogous. Writing $Z' := \text{Spec}(\mathcal{O}_{K,S})$ and $\overline{z_0} : \text{Spec}(\overline{K}) \to Z'$, the statements even kind of looks the same:

$$\pi_1^{\text{ét}}(Z', \overline{z_0})^{\text{ab}} \cong \text{Pic}(Z') \iff \pi_1(X', x_0)^{\text{ab}} \otimes \mathbb{R}/\mathbb{Z} \cong \text{Pic}^{\circ}(X').$$

Perhaps the most interesting thing here, however, is why these statements are different. Why the extra $\mathbb{R}/\mathbb{Z}$? Why π_1 instead of $\widehat{\pi_1} = \pi_1^{\text{ét}}$? Why Jac = Pic$^{\circ}$ instead of Pic? The answers to these questions are simple... I don't know. But we may find out in one of the bonus lectures!

2 Schemes and morphisms (Diego)

2.1 Some classical algebraic geometry

How much can we know about spaces from their functions? A simple example here is the space $\mathbb{C}^n$. Consider the set R of polynomial functions $\mathbb{C}^n \to \mathbb{C}$, which may be identified as $R = \mathbb{C}[x_1, \dots, x_n]$. Then for each point $p = (p_1, \dots, p_n) \in \mathbb{C}^n$, we may define the evaluation map $\mathrm{ev}_p : R \to \mathbb{C}$ given by $\mathrm{ev}_p(f) := f(p)$. Clearly this is a surjective morphism of $\mathbb{C}$-algebras, so we can think of this as a map quotienting out by the maximal ideal

$$\mathfrak{m}_p := \ker(\mathrm{ev}_p) = (x_1 - p_1, \dots, x_n - p_n).$$

In fact, all maximal ideals of R are of the form $\mathfrak{m}_q$ for some $q \in \mathbb{C}^n$, so this defines a bijection

$$\begin{array}{ccc} \mathbb{C}^n & \xleftrightarrow{\sim} & \mathrm{Spec}_{\mathfrak{m}}(R) \\ p & \longmapsto & \mathfrak{m}_p \end{array}$$

where $\mathrm{Spec}_{\mathfrak{m}}(R)$ denotes the *maximal spectrum* of R, that is, the set of maximal ideals of R. For each subset $U \subseteq \mathbb{C}^n$ and ideal $J \trianglelefteq R$, we may then extend this construction by defining

$$\mathbb{I}(U) := \bigcap_{p \in U} \mathfrak{m}_p, \quad \text{and} \quad \mathbb{V}(J) := \{p \in \mathbb{C}^n \mid \forall f \in J,\, f(p) = 0\}.$$

These two maps end up to be inclusion reversing, and thus define an antitone Galois connection

$$\left\{\begin{array}{c}\text{Subsets} \\ \text{of } \mathbb{C}^n\end{array}\right\} \underset{\mathbb{V}}{\overset{\mathbb{I}}{\rightleftarrows}} \left\{\begin{array}{c}\text{Ideals} \\ \text{of } R\end{array}\right\}$$

which restrict to inclusion reversing bijections between the *radical ideals* of R (i.e. those such that if $x^n \in J$ then $x \in J$), and the so-called *algebraic sets* (this is *Hilbert's Nullstellensatz*).

Definition 2.1. A subset $X \subseteq \mathbb{C}^n$ is called an *algebraic subset* if it is defined to be the common zero locus of a set of n-variable polynomials in $\mathbb{C}^n$.

We note that:

- $\mathbb{C}^n = \mathbb{V}(0)$, so $\mathbb{C}^n$ is an algebraic subset of itself;
- $\varnothing = \mathbb{V}(1)$, so $\varnothing$ is an algebraic subset of itself;
- as a consequence of *Hilbert's Basis Theorem* (i.e. that R is Noetherian), arbitrary intersections of algebraic subsets are algebraic; and
- as a consequence of the fact $\mathbb{V}(f) \cup \mathbb{V}(g) = \mathbb{V}(fg)$, finite unions of algebraic subsets are algebraic.

All of these properties say that the algebraic subsets define the closed sets of a topology on $\mathbb{C}^n$, which we call the *Zariski topology* $\mathcal{T}_{\mathrm{Zar}}$. That is, a subset of $\mathbb{C}^n$ is Zariski-closed if and only if it is algebraic.

Definition 2.2. When $\mathbb{C}^n$ (or equivalently, k^n for a field k) is endowed with the Zariski topology $\mathcal{T}_{\mathrm{Zar}}$, it is referred to as (complex) *affine n-space* $\mathbb{A}^n$. A Zariski closed subset of $\mathbb{A}^n$ endowed with the subspace topology is referred to as a (complex) *closed affine algebraic variety* (or just a *CAAV*).

We can check that a basis for the open sets of the Zariski topology is given by open sets of the form

$$D_f := \{p \in \mathbb{A}^n \mid f(p) \neq 0\},$$

which are huge for nonzero polynomials $f \in R$ (cf. Figure 1)! Because of this, CAAVs are very poorly behaved topological spaces (e.g. they're not Hausdorff), so continuous maps should not necessarily be our notion of a 'nice' map. We are usually more interested in polynomial functions on these spaces, which preserve this algebraic structure.

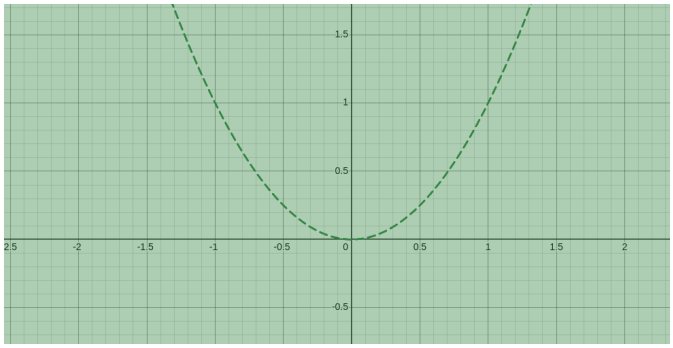


Figure 1: The basic open set D_{y-x^2} in $\mathbb{A}^2$ shaded in light green.

Example 2.3. Let $X = \mathbb{V}(x^2+y^2-1) \subseteq \mathbb{A}^2$ be the circle in the plane[1]. What would the polynomial functions $X \to \mathbb{A}^1$ even be? Well, these should be the restriction of a polynomial function $\mathbb{A}^2 \to \mathbb{A}^1$ to X, considered equal if and only if they agree at all points of X. The polynomial functions $\mathbb{A}^2 \to \mathbb{A}^1$ are just 2-variable complex polynomials, $\mathbb{C}[x,y]$, and two polynomials $f, g \in \mathbb{C}[x,y]$ agree on X if and only their difference $f-g$ is zero on X. The polynomials which are zero on X, by definition, are the elements of the vanishing ideal $\mathbb{I}(X) = (x^2+y^2-1)$, and so the polynomial functions $X \to \mathbb{A}^1$ may be identified with elements of the quotient $\mathbb{C}[x,y]/(x^2+y^2-1)$. ♢

Definition 2.4. For a closed affine algebraic variety $X \subseteq \mathbb{A}^n$, its *coordinate ring* $\mathcal{O}(X)$ is the ring of polynomial functions $X \to \mathbb{A}^1$, that is, it is the quotient ring

$$\mathcal{O}(X) := \frac{\mathbb{C}[x_1,\ldots,x_n]}{\mathbb{I}(X)}.$$

We note a few things about coordinate rings of CAAVs over $\mathbb{C}$:

- the coordinate ring $\mathcal{O}(X)$ is a $\mathbb{C}$-algebra, since constant functions are polynomial;
- since $\mathbb{I}(X)$ is radical and $\mathbb{C}[x_1,\ldots,x_n]$ is an ID, the quotient $\mathcal{O}(X)$ is *reduced* (it has no nilpotents);
- since $\mathbb{C}[x_1,\ldots,x_n]$ is Noetherian and finitely-generated as a $\mathbb{C}$-algebra, $\mathcal{O}(X)$ is finitely-generated as a $\mathbb{C}$-algebra.

This association of a coordinate ring to each CAAV over $\mathbb{C}$ may be seen to give an equivalence of categories

$$\left\{\begin{array}{c}\text{closed, affine,}\\ \text{algebraic varieties}\\ \text{over } \mathbb{C}\end{array}\right\} \overset{\simeq}{\longleftrightarrow} \left\{\begin{array}{c}\text{finitely generated,}\\ \text{reduced, commutative}\\ \mathbb{C}\text{-algebras}\end{array}\right\}^{\text{op}}$$

The fully-faithful property is pretty easy to see (our morphisms of CAAVs are just polynomial maps), but the essential surjectivity on objects is a bit more tricky. To show this we think back to the observation that at the beginning of the chapter that points of $\mathbb{A}^n$ are in bijection with maximal ideals of $\mathbb{C}[x_1,\ldots,x_n]$.

It is not too hard to show that this also holds of any complex CAAV X: that points of X are in bijection with maximal ideals of $\mathcal{O}(X)$. So if we endow $\text{Spec}_{\text{m}}(\mathcal{O}(X))$ with the Zariski topology, we can view the maximal spectrum *itself* as a CAAV.

[1] While we are usually considering things over $\mathbb{C}$, which has topological dimension 2, the algebraic dimension of affine n-space is n and so $\mathbb{A}^1$ is referred to as *the line*, $\mathbb{A}^2$ as *the plane*, etc.

To do this we only need to recall that the evaluation map ev_p may be seen as the quotient by $\mathfrak{m}_p$, so the 'evaluation' of a 'function' $f \in \mathcal{O}(X)$ at a 'point' $\mathfrak{m}_1 \in \mathrm{Spec}_{\mathfrak{m}}(\mathcal{O}(X))$ is defined to be the residue class $f \bmod \mathfrak{m}$. Translating the definition of the basic open sets of X into this language, a basis for the Zariski topology on $\mathrm{Spec}_{\mathfrak{m}}(\mathcal{O}(X))$ is given by the sets of the form

$$D_f := \{\mathfrak{m}_1 \in \mathrm{Spec}_{\mathfrak{m}}(\mathcal{O}(X)) \mid f \notin \mathfrak{m}_1\}.$$

Given a finitely generated, reduced, commutative $\mathbb{C}$-algebra A, its maximal spectrum $\mathrm{Spec}_{\mathfrak{m}}(A)$ under the Zariski topology may then be seen as a closed affine algebraic variety over $\mathbb{C}$ whose coordinate ring is A, giving essential surjectivity. To repeat,

a closed affine algebraic variety X **is** the maximal spectrum of its coordinate ring $\mathcal{O}(X)$:

$$X = \mathrm{Spec}_{\mathfrak{m}}(\mathcal{O}(X))$$

As for why this is functorial, the morphism of CAAVs corresponding to an algebra homomorphism $\varphi : A \to B$ is

$$\mathrm{Spec}_{\mathfrak{m}}(\varphi) : \mathrm{Spec}_{\mathfrak{m}}(B) \to \mathrm{Spec}_{\mathfrak{m}}(A), \qquad \mathfrak{m}_B \mapsto \phi^{-1}(\mathfrak{m}_B).$$

2.1.1 Why move to schemes?

- We want to create a similar theory that accepts *any* commutative ring, rather than exclusively finitely generated, reduced, commutative $\mathbb{C}$-algebras;
- Since algebraic subsets correspond to radical ideals, this theory does not see points with multiplicity/rings with nilpotence. For example, the equation $x^2 = 0$ in $\mathbb{A}^1$ should correspond to a point at zero *with multiplicity 2*, but $\mathbb{V}(x^2) = \mathbb{V}(\sqrt{(x^2)}) = \mathbb{V}(x)$ which only sees that there is a point at zero and forgets the multiplicity;
- While $\mathrm{Spec}_{\mathfrak{m}}$ is functorial for our FGRC $\mathbb{C}$-algebras, it fails to be functorial in general (preimages of maximal ideals under general ring homomorphisms are rarely maximal, e.g. (0) in $\mathbb{Z} \to \mathbb{Q}$).

2.2 Affine schemes

In much the same way that the theory of varieties revolves around the maximal spectrum, the theory of schemes is based on the *spectrum* of a commutative ring R. This is the set of *prime* ideals of R, which we also endow with the Zariski topology

$$\mathrm{Spec}(R) = \{\mathfrak{p} \trianglelefteq R \text{ prime}\}, \qquad D_f = \{\mathfrak{p} \in \mathrm{Spec}(R) \mid f \notin \mathfrak{p}\}, \qquad \mathcal{T}_{\mathrm{Zar}} = \langle D_f \mid f \in R \rangle.$$

For simplicity, we will assume from now on that all rings R are integral domains unless otherwise specified, but bear in mind that all of this can be done over non-integral domains (the definition of localisation does change in this case though).

Definition 2.5. An *affine scheme* is a pair $(X, \mathcal{O}_X)$, where $X = \mathrm{Spec}(R)$ for some commutative ring R, and $\mathcal{O}_X$ is a sheaf[a] on X (called the *structure sheaf*) defined on the basic Zariski-open sets D_f to be the localisation of R at f

$$\mathcal{O}_X(D_f) := R_f = R\left[\frac{1}{f}\right],$$

i.e. those elements of the field of fractions $\mathrm{Frac}(R)$ of the form g/f^n, where $g \in R$ and $n \in \mathbb{Z}$. The restriction morphisms $\mathrm{res} : \mathcal{O}_X(D_f) \to \mathcal{O}_X(D_{fg})$ is the usual inclusion map. The ring $\mathcal{O}(X) := \mathcal{O}_X(X)$ of global sections is called the *coordinate ring* of X.

[a] If you don't know what a sheaf is, I've included a short appendix on everything you'll need to know about them.

We can check that the stalks of $\mathcal{O}_X$ are given by

$$\mathcal{O}_{X,\mathfrak{p}} = \operatorname*{colim}_{U \ni \mathfrak{p}} \mathcal{O}_X(U) = R_{\mathfrak{p}},$$

the localisation of R at $\mathfrak{p}$, i.e. those elements of $\mathrm{Frac}(R)$ whose denominator is not in $\mathfrak{p}$. Since $R_{\mathfrak{p}}$ is local, affine schemes are locally ringed spaces. If a locally ringed space $(X, \mathcal{O})$ is isomorphic to an affine scheme, we will call it an affine scheme also.

As before, for each $\mathfrak{p} \in \operatorname{Spec}(R)$ we define the *evaluation map* $\operatorname{ev}_\mathfrak{p}$ to be the quotient map with kernel $\mathfrak{p}$, although now it is valued in the *residue field* $k(\mathfrak{p}) := \operatorname{Frac}(R/\mathfrak{p}) = R_\mathfrak{p}/\mathfrak{p}R_\mathfrak{p}$. This allows us to define our $\mathbb{V}$ and $\mathbb{I}$ functions as before:

$$\mathbb{V}(J) := \{\mathfrak{p} \in \operatorname{Spec}(R) \mid \operatorname{ev}_\mathfrak{p}(f) = 0,\ \forall f \in J\} = \{\mathfrak{p} \in \operatorname{Spec}(R) \mid J \subseteq \mathfrak{p}\},$$
$$\mathbb{I}(U) := \{f \in R \mid \operatorname{ev}_\mathfrak{p}(f) = 0,\ \forall \mathfrak{p} \in U\} = \bigcap_{\mathfrak{p} \in U} \mathfrak{p},$$

where $J \trianglelefteq R$ is an ideal and $U \subseteq \operatorname{Spec}(R)$ is a subset. Notably, we see that:

- the Zariski-closure $\overline{U}$ of a subset $U \subseteq \operatorname{Spec}(R)$ is exactly $\overline{U} = \mathbb{V}\mathbb{I}(U)$;
- the correspondence theorem ensures that $\mathbb{V}(J) \subseteq \operatorname{Spec}(R)$ may be identified with the affine scheme $\operatorname{Spec}(R/J)$, so the Zariski-closed subsets are the spectra of quotients of R (and $\varnothing$);
- properties of localisation ensure that $D_f \subseteq \operatorname{Spec}(R)$ may be identified with the affine scheme $\operatorname{Spec}(R_f)$, so the basic Zariski-open subsets are the spectra of localisations of R.

Example 2.6. (i) The affine scheme $\operatorname{Spec}(R[x_1, \ldots, x_n])$ is again called *affine n-space* $\mathbb{A}^n$, and to specify the ring R we say that it is *defined over* R. This is sometimes denoted $\mathbb{A}^n_R$ or $\mathbb{A}^n_{/R}$.

(ii) While we have seen that the closed affine algebraic variety $\operatorname{Spec}_\mathfrak{m}(\mathbb{C}[x, y])$ may be identified with $\mathbb{C}^n$ under the Zariski topology, we instead look at the prime spectrum $\operatorname{Spec}(\mathbb{C}[x, y])$. As a set, this decomposes as

$$\operatorname{Spec}(\mathbb{C}[x, y]) = \operatorname{Spec}_\mathfrak{m}(\mathbb{C}[x, y]) \cup \{(0)\} \cup \{(f) \mid f \in \mathbb{C}[x, y] \text{ irr.}\}.$$

These different categories of points are distinguished by the Zariski closures of their corresponding singletons, which is exactly $\mathbb{V}\mathbb{I}(\{\mathfrak{p}\}) = \mathbb{V}(\mathfrak{p})$.

- For a point $\mathfrak{m}_1 \in \operatorname{Spec}_\mathfrak{m}(\mathbb{C}[x, y])$, i.e. a maximal ideal of $\mathbb{C}[x, y]$, we see that $\overline{\{\mathfrak{m}_1\}} = \mathbb{V}(\mathfrak{m}_1)$ is the set of prime ideals containing $\mathfrak{m}_1$, which is exactly the singleton $\{\mathfrak{m}_1\}$ since $\mathfrak{m}_1$ is maximal. For this reason, maximal ideals of $\mathbb{C}[x, y]$ correspond to *closed points* of $\operatorname{Spec}(\mathbb{C}[x, y])$ (and this works for general rings!);
- The point $(0) \in \operatorname{Spec}(\mathbb{C}[x, y])$, i.e. the zero ideal, has $\overline{\{(0)\}} = \mathbb{V}(0) = \operatorname{Spec}(\mathbb{C}[x, y])$. For this reason, the point (0) is called a *dense point* or a *generic point* of $\operatorname{Spec}(\mathbb{C}[x, y])$ (and again, this works for general rings);
- A point $(f) \in \operatorname{Spec}(\mathbb{C}[x, y])$ with $f \in \mathbb{C}[x, y]$ irreducible has $\overline{\{(f)\}} = \mathbb{V}(f) = \operatorname{Spec}(R/(f))$, which is a closed affine subscheme (i.e. a Zariski closed subset). Such points are thus the generic points of closed affine subschemes of $\operatorname{Spec}(\mathbb{C}[x, y])$.

So the points of $\operatorname{Spec}(\mathbb{C}[x, y])$ may be thought of as

$$\operatorname{Spec}(\mathbb{C}[x, y]) = \begin{Bmatrix}\text{closed}\\ \text{points}\end{Bmatrix} \cup \begin{Bmatrix}\text{generic}\\ \text{point}\end{Bmatrix} \cup \begin{Bmatrix}\text{generic points of}\\ \text{closed subschemes}\end{Bmatrix}$$

We may think of generic points as a sort of 'codimension 0 point', which is close to everything.

(iii) The affine scheme $\operatorname{Spec}(\mathbb{C}[x]/(x^2))$ decomposes as

$$\operatorname{Spec}\left(\frac{\mathbb{C}[x]}{(x^2)}\right) = \{(x)\} \cup \{(0)\}$$

which is just the one closed point, along with the generic point. This is seen to be different to $\operatorname{Spec}(\mathbb{C}[x]/(x)) = \operatorname{Spec}(\mathbb{C})$, which consists solely of a generic point. So our notion is tracking multiplicity correctly!

(iv) One interesting example that may motivate something later: if we consider the prime spectrum of $\mathbb{Z}$, this may be visualised as a long line (corresponding to the generic point) connecting infinitely many points (corresponding to the closed points, the ideals $p\mathbb{Z}$), as in Figure 2.

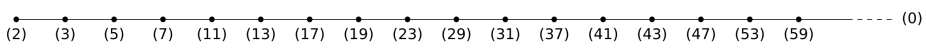


Figure 2: $\operatorname{Spec}(\mathbb{Z})$, per the LMFDB.

The inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}[\sqrt{3}]$ gives us a corresponding surjective morphism $\mathrm{Spec}(\mathbb{Z}[\sqrt{3}]) \twoheadrightarrow \mathrm{Spec}(\mathbb{Z})$ which may be visualised as in Figure 3, where the vertical columns are the fibres of the morphism.

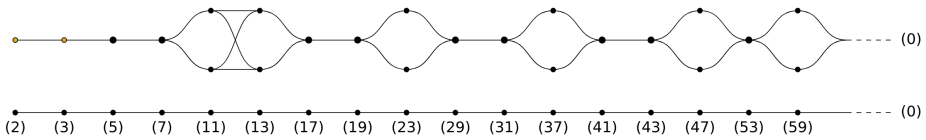


Figure 3: $\mathrm{Spec}(\mathbb{Z}[\sqrt{3}]) \twoheadrightarrow \mathrm{Spec}(\mathbb{Z})$, per the LMFDB.

The behaviour of different points fall into three classes, all depicted differently in this diagram:

- **A single, black point** above a point (p) of $\mathrm{Spec}(\mathbb{Z})$ corresponds to the ideal $p\mathbb{Z}[\sqrt{3}]$ being a prime ideal of $\mathbb{Z}[\sqrt{3}]$, as we see with e.g. (17). This point $17\mathbb{Z}[\sqrt{3}]$ of $\mathrm{Spec}(\mathbb{Z}[\sqrt{3}])$ is a *reduced point*, i.e. a point with multiplicity 1. In the number theory lingo, we say that 17 is *inert* in the extension $\mathbb{Q}(\sqrt{3})/\mathbb{Q}$;
- **Two different black points** above a point (p) of $\mathrm{Spec}(\mathbb{Z})$ corresponds to the ideal $p\mathbb{Z}[\sqrt{3}]$ decomposing as a product of two distinct prime ideals $\mathfrak{P}_1, \mathfrak{P}_2$ in $\mathbb{Z}[\sqrt{3}]$ each with multiplicity 1, as we see with e.g. (11) decomposing in $\mathbb{Z}[\sqrt{3}]$ as $(1-2\sqrt{3})(1+2\sqrt{3})$. Both of these points $\mathfrak{P}_1, \mathfrak{P}_2$ above (p) are again reduced points of $\mathrm{Spec}(\mathbb{Z}[\sqrt{3}])$. In the number theory lingo, we say that 11 is *completely split* in $\mathbb{Q}(\sqrt{3})/\mathbb{Q}$, and we say that $(1-2\sqrt{3})$ and $(1+2\sqrt{3})$ are the primes *above* 11;
- **A single, yellow point** above a point (p) of $\mathrm{Spec}(\mathbb{Z})$ corresponds to the ideal $p\mathbb{Z}[\sqrt{3}]$ being the square[2] of a single prime ideal $\mathfrak{P}$ in $\mathbb{Z}[\sqrt{3}]$, as we see with e.g. $3\mathbb{Z}[\sqrt{3}] = (\sqrt{3})^2$. This point $(\sqrt{3})$ is a *non-reduced point* of multiplicity 2. In the number theory lingo, we see that 3 is *totally ramified* in $\mathbb{Q}(\sqrt{3})/\mathbb{Q}$, and we say that $(\sqrt{3})$ is the prime above 3.

♢

By construction, we get a similar equivalence of categories

$$\begin{array}{ccc} \mathbf{CRing} & \xleftarrow{\simeq} & \mathbf{AffSch}^{\mathrm{op}} \\ R & \longmapsto & (\mathrm{Spec}(R), \mathcal{O}_{\mathrm{Spec}(R)}) \end{array}$$

between the categories of commutative rings and the (opposite) category of affine schemes. By definition of affine schemes, this map is clearly essentially surjective. The more difficult thing to prove here is that all morphisms of affine schemes are induced by ring homomorphisms. We omit this proof.

2.2.1 A quick note on abstract schemes

While we have defined the notion of an affine scheme, there is a more general notion of a scheme which is a strictly larger category. Very roughly, affine schemes are to schemes what $\mathbb{C}^n$ is the a complex manifold: a scheme should be a space that locally looks like an affine scheme, or one that results from gluing together affine schemes.

Definition 2.7. A locally ringed space $(X, \mathcal{O})$ is called a *scheme* if there is an open cover $\{U_i\}_{i \in I}$ of X such that each U_i, as a locally ringed space, is isomorphic to an affine scheme $\mathrm{Spec}(R_i)$. A *morphism of schemes* is then a map of locally ringed spaces between schemes.

Remark. A quick etymological note: the reason why we use the somewhat-ominous word *scheme* for these things is due to a slight mistranslation. Originally called *schémas* in French by Chevalley, this should have been translated to *schema*, i.e. a universally applicable outline of a general concept (per Wiktionary). Indeed, the word *schema* is much more apt, since they allow us to define sets of points via a general rule for every R-algebra, as we shall see.

[2] ...or more generally, a power of a single prime ideal.

Example 2.8. (i) Let $X = \mathbb{A}^2$ be affine 2-space, say over $\mathbb{C}$. Then $U := \mathbb{A}^2 \setminus \{(0,0)\}$ defines a Zariski-open subset, which is covered by the Zariski opens

$$U_1 = \operatorname{Spec}\left(\frac{\mathbb{C}[x,y,z]}{(xz-1)}\right) \qquad \text{and} \qquad U_2 = \operatorname{Spec}\left(\frac{\mathbb{C}[x,y,z]}{(yz-1)}\right).$$

Since U has an open cover by affine schemes, it is itself a scheme. However, since every regular function on U (i.e. locally rational) extends uniquely to a regular function on X, the map $\mathcal{O}(X) \to \mathcal{O}(U)$ induced by the inclusion $U \hookrightarrow X$ is an isomorphism. But $U \hookrightarrow X$ is not itself an isomorphism, meaning that U cannot be affine by the equivalence of categories (only isomorphisms should induce isomorphisms).

(ii) **Proj construction:** If $R = \bigoplus_{i\geq 0} R_i$ is a commutative graded ring, we define the ideal $R_+ = \bigoplus_{i>0} R_i$ of positive-degree elements. An ideal $J \trianglelefteq R$ is *homogeneous* if it is generated by homogeneous elements (not necessarily of the same degree). Let $Z := \operatorname{Proj}(R)$ denote the set of prime homogeneous ideals of R which do not contain R_+. Then we define the *projective Zariski topology* $\mathcal{T}^{\mathbb{P}}_{\text{Zar}}$ to be given by the basic opens $D_0(f) = \{\mathfrak{p} \in Z \mid f \notin \mathfrak{p}\}$, where $f \in R_d$ is a homogeneous element of R.

We then see that the open set $D_0(f)$ may be identified with $\operatorname{Proj}(R_f)$, which may further be identified with $\operatorname{Spec}(R_{f,0})$ as a topological space, where $R_{f,0}$ is the subring of degree-zero homogeneous elements of R_f. We may then define the structure sheaf $\mathcal{O}_Z$ on Z by identifying $D_0(f) = \operatorname{Spec}(R_{f,0})$ as *affine schemes*, so $\mathcal{O}_Z(D_0(f)) = R_{f,0}$. By construction, we see that $\operatorname{Proj}(R)$ is a scheme, but it is not affine.

(iiia) As a specific example of the Proj construction, we define *projective n-space* $\mathbb{P}^n$ over R to be the scheme

$$\mathbb{P}^n := \operatorname{Proj}(R[x_0,\dots,x_n]),$$

where $R[x_0,\dots,x_n]$ is given its canonical grading by degree. When $R = \mathbb{R}$ or $R = \mathbb{C}$, this agrees with the classical notion of $\mathbb{RP}^n$ or $\mathbb{CP}^n$, insomuchas the closed points naturally identify with these manifolds when endowed with the Euclidean topology. As with affine n-space, this is sometimes denoted $\mathbb{P}^n_R$ or $\mathbb{P}^n_{/R}$.

(iiib) Alternatively, we can define $\mathbb{P}^n$ via gluing: in the case $n = 1$, we may two copies of $R[y]$ as subrings of $R[t,t^{-1}]$, namely $R[t]$ and $R[t^{-1}]$. Letting $X = \operatorname{Spec}(R[u]), Y = \operatorname{Spec}(R[v])$ be two copies of the affine line, we then get two maps

$$\varphi_* : \operatorname{Spec}(R[t,t^{-1}]) \to X, \qquad \psi_* : \operatorname{Spec}(R[t,t^{-1}]) \to Y,$$

induced by these two maps $\varphi : R[u] \to R[t,t^{-1}]; u \mapsto t$ and $\psi : R[v] \to R[t,t^{-1}]; v \mapsto t^{-1}$, and we can naturally identify their images as schemes. Gluing along this identification, we may define

$$\mathbb{P}^1 := \frac{X \sqcup Y}{\varphi_*(x) \sim \psi_*(x)}$$

and then since structure sheaves agree on $\operatorname{im}(\varphi_*)$ and $\operatorname{im}(\psi_*)$, the structure sheaf of $\mathbb{P}^1$ is well-defined by picking affine charts. We similarly define $\mathbb{P}^n$ by defining polynomial rings

$$A_i = R[u_{i,0},\dots,\widehat{u_{i,i}},\dots,u_{i,n}], \qquad B_{i,j} = R[t_{i,1},\dots,\widehat{t_{i,i}},\dots,t_{i,n},t_{i,j}^{-1}],$$

for each $j \neq i$, where the hat denotes that $u_{i,i}$ and $t_{i,i}$ are omitted. We then get affine charts $X_i := \operatorname{Spec}(A_i) \cong \mathbb{A}^n$ with open immersions $\operatorname{Spec}(B_{i,j}) \to X_i$ and $\operatorname{Spec}(B_{i,j}) \to X_j$ induced by the maps

$$\begin{aligned} \varphi_i : A_i &\to B_{i,j} \\ u_{i,k} &\mapsto t_{i,k} \end{aligned} \qquad\qquad \begin{aligned} \psi_i : A_j &\to B_{i,j} \\ u_{j,i} &\mapsto t_{i,j}^{-1} \\ u_{j,k} &\mapsto t_{i,k}t_{i,j}^{-1} \ (k \neq i,j) \end{aligned}$$

We then define

$$\mathbb{P}^n = \frac{\coprod_{i=0}^n \operatorname{Spec}(A_i)}{\varphi_{i,*}(x) \sim \psi_{i,*}(x)}$$

Once again, this also defines the structure sheaf on $\mathbb{P}^n$. ♢

One very important notion is that of a scheme being *above* another scheme.

Definition 2.9. If S is a scheme, then an *S-scheme* is a scheme X equipped with a morphism $\pi : X \to S$. If R is a ring, then we may call a $\mathrm{Spec}(R)$-scheme simply an *R-scheme*. A *morphism* $f : (X_1, \pi_1) \to (X_2, \pi_2)$ of S-schemes is a morphism of schemes that commutes with the maps $\pi_i : X_i \to S$, i.e. such that $\pi_2 \circ f = \pi_1$.

Example 2.10. (i) Any affine scheme $\mathrm{Spec}(R)$ is a $\mathbb{Z}$-scheme in a unique way, since any commutative ring admits a unique homomorphism $\mathbb{Z} \to R$;

(ii) Affine n-space $\mathbb{A}^n$ over R is an R-scheme because $R[x_1, \ldots, x_n]$ is naturally an R-algebra;

(iii) Any extension L/K of fields makes $\mathrm{Spec}(L)$ a K-scheme;

(iv) Any extension L/K of number fields makes $\mathrm{Spec}(\mathcal{O}_L)$ an $\mathcal{O}_K$-scheme, and if S is a set of primes of $\mathcal{O}_K$ then $\mathrm{Spec}(\mathcal{O}_{L,S})$ becomes an $\mathcal{O}_{K,S}$-scheme (where $\mathcal{O}_{K,S}$ denotes the S-integers of K and similarly for L);

(v) The abstract scheme $U := \mathbb{A}^2 \setminus \{(0,0)\}$ from Example 2.8(i) has a canonical projection map to the affine scheme $S := \mathbb{A}^1 \setminus \{0\} = \mathrm{Spec}(\mathbb{C}[x,y]/(xy-1))$, making U an S-scheme.

♢

2.3 Fibre products and base change

Along with the notion of schemes and morphisms, we have a few notable operations we can do on schemes. The first and foremost is the notion of *fibre products*.

Definition 2.11. If $\varphi : X \to Z$ and $\psi : Y \to Z$ are two morphisms of schemes, the *fibre product* $X \times_Z Y$ is the limit of the diagram $X \to Z \leftarrow Y$, that is, it is universal with respect to the commutative diagram

$$\begin{array}{ccc} X \times_Z Y & \dashrightarrow & Y \\ \vdots & & \downarrow \psi \\ X & \xrightarrow{\varphi} & Z \end{array}$$

An important special case of the fibre product is the *base change* $X_A := X \times_{\mathrm{Spec}(R)} \mathrm{Spec}(A)$, where X is an R-scheme and A is a commutative R-algebra.

Example 2.12. (i) If $X := \mathbb{A}^n_R$ is affine n-space over a commutative ring R and A is a commutative R-algebra, then the base change X_A is exactly affine n-space over A;

(ii) Similarly, if $X := \mathbb{P}^n_R$ is projective n-space over a commutative ring R and A is a commutative R-algebra, then the base change X_A is exactly projective n-space over A.

♢

Proposition 2.13. If $X = \mathrm{Spec}(A), Y = \mathrm{Spec}(B), Z = \mathrm{Spec}(R)$ are affine schemes with morphism $\varphi : X \to Z$ and $\psi : Y \to Z$ (making A and B commutative R-algebras), then their fibre product is $X \times_Z Y = \mathrm{Spec}(A \otimes_R B)$.

Proof. By the equivalence of categories $\mathbf{CRing}^{\mathrm{op}} \simeq \mathbf{AffSch}$, the pullback in **AffSch** corresponds to a pushout in **CRing**, which are given exactly by tensor products. So this is true as a pullback in **AffSch**. It suffices to note that the inclusion $\mathbf{AffSch} \hookrightarrow \mathbf{Sch}$ preserves limits because it is right-adjoint to $X \mapsto \mathrm{Spec}(\mathcal{O}_X(X))$. □

Example 2.14. Consider affine n- and m-space $\mathbb{A}^n, \mathbb{A}^m$ over $\mathbb{C}$. Let $C = \mathrm{Spec}(\mathbb{C})$, which we may view as a point onto which both $\mathbb{A}^n$ and $\mathbb{A}^m$ project. The fibre product $\mathbb{A}^n \times_C \mathbb{A}^m$ may then be compute using Proposition 2.13:

$$\mathbb{A}^1 \times_C \mathbb{A}^1 = \mathrm{Spec}(\mathbb{C}[x_1, \ldots, x_n] \otimes_{\mathbb{C}} \mathbb{C}[y_1, \ldots, y_m]) = \mathrm{Spec}(\mathbb{C}[x_1, \ldots, x_n, y_1, \ldots, y_m]) = \mathbb{A}^{n+m}.$$ ♢

2.4 Some special families of morphisms

Recall that we defined a morphism from a scheme X to another scheme Y to be a morphism $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ of the underlying locally ringed spaces. What does this actually mean for these functions? Well, continuity with respect to the Zariski topology should mean that the preimage of a subscheme is a subscheme, so a continuous function $f : X \to Y$ should be one that respects the "subscheme" heirarchy. The existence of the corresponding sheaf morphism $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$ just tells us that this also respects the underlying *algebraic* structure.

In the affine case, the equivalence of categories $\mathbf{CRing}^{\mathrm{op}} \simeq \mathbf{AffSch}$ tells us that it is enough to respect the algebraic structure (i.e. the global component $f^\#_Y : \mathcal{O}(Y) \to \mathcal{O}(X)$ of $f^\#$ fully determines the map f). More generally, morphisms of schemes restrict to morphisms of affine charts, so we can think of a morphism of a general scheme (in a very imprecise way) to be a bunch of ring homormophisms glued together.

As such, the usual families of ring homomorphisms (finite type, flat, etc.) may be extended to families of *scheme* morphisms. In this section, we'll define all of the families of morphisms we need.

2.4.1 Finiteness criteria

First on the chopping block are the so-called finite-type morphisms. Recall that a homomorphism $f : A \to B$ of commutative rings is said to be of finite type if it realises B as an A-algebra of finite-type (i.e. it is isomorphic to a quotient of some $A[x_1, \dots, x_n]$).

Definition 2.15. A morphism $f : \mathrm{Spec}(S) \to \mathrm{Spec}(R)$ of affine schemes is said to be of *finite type* if the associated ring homomorphism $f^\# : R \to S$ is finite-type. We say that a morphism $f : X \to Y$ of abstract schemes is *locally of finite type* if for every $x \in X$, there is an affine chart $U \ni x$ of X and an affine chart $V \supseteq f[U]$ of Y such that the restriction $f|_{U,V} : U \to V$ is of finite type.

Example 2.16. (i) Let $n \leq m$ and consider affine n- and m-space $\mathbb{A}^n, \mathbb{A}^m$ over $\mathbb{C}$. The map

$$f : \mathbb{A}^m \twoheadrightarrow \mathbb{A}^n; (x_0, \dots, x_m) \mapsto (x_0, \dots, x_n)$$

induced by inclusion $f^\# : \mathcal{O}(\mathbb{A}^n) \hookrightarrow \mathcal{O}(\mathbb{A}^m)$ is of finite type, since the associated homomorphism $f^\#$ trivially realises $\mathbb{C}[x_1, \dots, x_m]$ as a $\mathbb{C}[x_1, \dots, x_n]$-module;

(ii) Now consider the affine scheme $\mathbb{A}^1 \setminus \{0\} = \mathrm{Spec}(\mathbb{C}[x, x^{-1}])$ over $\mathbb{C}$. The map $w : \mathbb{A}^1 \setminus \{0\} \twoheadrightarrow \mathbb{A}^1 \setminus \{0\}; z \mapsto z^n$ with $w^\# : \mathbb{C}[x, x^{-1}] \hookrightarrow \mathbb{C}[y, y^{-1}]; x \mapsto y^n$, clearly realising $\mathbb{C}[y, y^{-1}]$ as a quotient of $\mathbb{C}[x, x^{-1}][u, v]$, namely

$$\mathbb{C}[y, y^{-1}] \cong \frac{\mathbb{C}[x, x^{-1}][u, v]}{(u^n - x, v^n - x^{-1}, uv - 1)},$$

clearly by $y \mapsto u, y^{-1} \mapsto v = u^{-1}$;

(iii) If L/K is a finite extension of number fields, then inclusion $K \hookrightarrow L$ induces the finite type morphism $\mathrm{Spec}(L) \twoheadrightarrow \mathrm{Spec}(K)$, and similarly we get finite type morphisms $\mathrm{Spec}(\mathcal{O}_L) \twoheadrightarrow \mathrm{Spec}(\mathcal{O}_K)$, and similarly for S-integers.

♢

There is a slightly stronger notion than finite-type called *finite-presentation*. A homomorphism $f : A \to B$ of rings is of *finite presentation* if it makes B a finitely *finitely presented* A-algebra, i.e. it is a quotient of some $A[x_1, \dots, x_n]$ by some finitely generated ideal $(f_1, \dots, f_m)$.

Definition 2.17. A morphism $f : \mathrm{Spec}(S) \to \mathrm{Spec}(R)$ of affine schemes is said to be of *finite presentation* if the associated ring homomorphism $f^\# : R \to S$ is finite-presentation. We say that a morphism $f : X \to Y$ of abstract schemes is *locally of finite presentation* if for every $x \in X$, there is an affine chart $U \ni x$ of X and an affine chart $V \supseteq f[U]$ of Y such that the restriction $f|_{U,V} : U \to V$ is of finite presentation.

Note that if A is Noetherian then so too is $A[x_1, \dots, x_n]$, so a morphism $f : A \to B$ is of finite presentation if and only if it is of finite type. Consequently, any (locally) finite-type morphism $X \to \mathrm{Spec}(A)$ is also (locally) of finite presentation.

Example 2.18. (i) All of the examples from Example 2.16 are of finite presentation, since all fields and Dedekind domains are Noetherian;

(ii) **Non-example**: if k is a field, then the polynomial ring $A := k[x_1, x_2, \ldots]$ in countably many variables is not Noetherian. The evaluation map $A \to k[x]$ sending $x_1 \mapsto x$ and $x_n \mapsto 0$ for $n > 1$ realises $k[x]$ as a finite-type A-algebra (in fact, with zero generators!), but the kernel $(x_2, x_3, x_4, \ldots)$ is not finitely generated over A. Thus the map $\mathbb{A}^1_k \to \mathrm{Spec}(A)$ induced by this is of finite type but not finite presentation.

♢

Most of the rings we work over will be Noetherian, so the distinction between morphisms of locally finite-type and of locally finite-presentation will be one that we will not pay much attention to.

2.4.2 Relative differentials

To define the other properties we need, we will first have to talk about *relative differentials* of ring homomorphisms. The idea comes from differential geometry: the differential forms on a manifold tell us a lot about that manifold's geometry. However, in algebraic geometry we have no analytic structure with which to define differential forms, but we can define derivations. The relative differentials then form some universal derivation, and thus encapsulate the idea of a differential of a function.

Definition 2.19. Let R be a commutative ring and S a commutative R-algebra. Then an *R-derivation* on S is an R-linear map $d : S \to M$ to an S-module M satisfying the *Leibniz rule*:

$$d(fg) = f \cdot dg + g \cdot df.$$

Proposition 2.20. Let $\varphi : R \to S$ be a homomorphism of commutative rings, making S an R-algebra. Let $\widetilde{\Omega}$ denote the free S-module generated by the symbols $\{ds \mid s \in S\}$, and $N \leq \widetilde{\Omega}$ the submodule given by

$$N := \langle d(\varphi r),\ d(s+t) - ds - dt,\ d(st) - sdt - tds \mid r \in R, s, t \in S \rangle.$$

Then the quotient module $\Omega^1_{S|R} := \widetilde{\Omega}/N$ admits a canonical R-derivation $d : S \to \Omega^1_{S|R}$ sending each $s \in S$ to $ds \in \Omega^1_{S|R}$. Such an R-derivation $d : S \to \Omega^1_{S|R}$ is *universal*, in that every other R-derivation $\delta : S \to M$ factors in the the form $\delta = \psi \circ d$, for $\psi : \Omega^1_{S|R} \to M$ a homomorphism of S-modules. Morover, if $\{s_1, \ldots, s_m\}$ generate S as an R-algebra, then $\Omega^1_{S|R}$ is generated by $\{ds_1, \ldots, ds_m\}$ as an S-module.

Proof. ∗ **Exc**. Prove this! (it's not too hard, most of the work is in the construction itself). □

Definition 2.21. The module $\Omega^1_{S|R}$, equipped with its universal R-derivation $d : S \to \Omega^1_{S|R}$, is called the module of *relative differentials*, or *Kähler differentials*, of S over R.

Example 2.22. (i) Let $A = \mathbb{C}[x_1, \ldots, x_n]$, considered a $\mathbb{C}$-algebra. Then since $x_1, \ldots, x_n$ generate A as a $\mathbb{C}$-algebra we can conclude, from $\mathbb{C}$-linearity of $\mathbb{C}$-derivations as well as the Leibniz rule, that $\Omega^1_{A|\mathbb{C}}$ is generated by the differentials $dx_1, \ldots, dx_n$. Thus, every Kähler differential $\omega \in \Omega^1_{A|\mathbb{C}}$ is of the form

$$\omega = f_1 dx_1 + \cdots + f_n dx_n, \qquad f_i \in A.$$

(ii) Suppose that L/K is a finite separable extension of fields. Then by the primitive element theorem, $L = K(\alpha)$ for some $\alpha \in L$. Thus, $\Omega^1_{L|K}$ is generated by $d\alpha$. Let $m = X^n + \sum_{i=0}^{n-1} a_i x^i \in K[x]$ denote the minimal polynomial of α over K, which is notably a separable irreducible polynomial so shares no roots with its formal derivative f'. Then

$$m(\alpha) = \alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0.$$

Applying the derivation d to both sides gives

$$0 = (n\alpha^{n-1} + (n-1)a_{n-1}\alpha^{n-2} + \cdots + 2a_2\alpha + a_1)d\alpha = f'(\alpha)d\alpha.$$

But $f'(\alpha) \neq 0$ since f is separable irreducible, and thus we conclude that $d\alpha = 0$, hence $\Omega^1_{L|K} = 0$. ♢

Proposition 2.23. Let S be a commutative R-algebra (for R a commutative ring) and consider the universal derivation $d : S \to \Omega^1_{S|R}$. If $p \in R[t]$ is a polynomial over R and $\alpha \in S$, then $d(p(\alpha)) = p'(\alpha)d\alpha$.

Proof. We may check that the usual power law $d(\alpha^n) = n\alpha^{n-1}d\alpha$ works by repeatedly using the Leibniz rule. WLOG we may write $p = \sum_{i=0}^n a_i t^i$, then we compute that

$$\begin{aligned} d(p(\alpha)) &= d\left(a_n\alpha^n + a_{n-1}\alpha^{n-1} + \cdots a_1\alpha + a_0\right) \\ &= a_n d(\alpha^n) + a_{n-1}d(\alpha^{n-1}) + \cdots + a_1 d\alpha + d(a_0) \\ &= n a_n \alpha^{n-1} d\alpha + (n-1)a_{n-1}\alpha^{n-2}d\alpha + \cdots + 2a_2\alpha d\alpha + a_1 d\alpha \\ &= p'(\alpha)d\alpha. \end{aligned}$$

□

We can then extend this notion for schemes using our notion of $\mathcal{O}_X$-modules (see §A.4.1).

Definition 2.24. If $f : Y \to X$ is a morphism of affine schemes, then the *sheaf of relative differentials* $\Omega^1_{Y|X}$ is the $\mathcal{O}_X$-module on X whose sections are given by $\Omega^1_{Y|X}(U) := \Omega^1_{f_*\mathcal{O}_Y(U)|\mathcal{O}_X(U)}$, with respect to the map $f^\#_U : \mathcal{O}_X(U) \to f_*\mathcal{O}_Y(U)$. For a morphism of general schemes $f : Y \to X$, we may define the sheaf $\Omega^1_{Y|X}$ by declaring it to be given by the above on the affine charts of X, Y.

Example 2.25. (i) Consider the $\mathbb{C}$-algebra homomorphim $\pi_n : \mathbb{C}[x, x^{-1}] \hookrightarrow \mathbb{C}[t, t^{-1}]$ given by $x \mapsto t^n$. This corresponds to the morphism Π_n of affine schemes given by

$$\Pi_n : \mathbb{G}'_{\mathrm{m}} \twoheadrightarrow \mathbb{G}_{\mathrm{m}}; \quad z \mapsto z^n,$$

where $\mathbb{G}_{\mathrm{m}} = \mathbb{G}'_{\mathrm{m}} := \mathrm{Spec}(\mathbb{C}[x, x^{-1}]) = \mathbb{A}^1_{\mathbb{C}} \setminus \{0\}$ are two copies of what is often called the *multiplicative group scheme* (we will see what this means later). We will then compute the sections $\Omega^1_{\mathbb{G}'_{\mathrm{m}}|\mathbb{G}_{\mathrm{m}}}$ on the basic open set $D_f \subseteq \mathbb{G}_{\mathrm{m}}$ for some $f \in \mathbb{C}[x, x^{-1}]$. By definition $\Omega^1_{\mathbb{G}'_{\mathrm{m}}|\mathbb{G}_{\mathrm{m}}}(D_f)$ is the module of relative differentials of

$$B := \Pi_{n,*}\mathcal{O}_Y(D_f) = \mathbb{C}[t, t^{-1}]_{\pi_n(f)} \quad \text{over} \quad A := \mathcal{O}_X(D_f) = \mathbb{C}[x, x^{-1}]_f.$$

The monomorphism π_n identifies $A = \mathbb{C}[t^n, t^{-n}]_{\pi_n(f)} \subseteq B$, and thus we may view $B = A[x^{1/n}, x^{-1/n}]$. Writing $s_1 = x^{1/n}$ and $s_2 = x^{-1/n}$, we thus see that $\Omega^1_{\mathbb{G}'_{\mathrm{m}}|\mathbb{G}_{\mathrm{m}}}(D_f)$ is generated as a B-module by $\{ds_1, ds_2\}$. Applying the Leibniz rule to the equation $s_1 s_2 = 1$, we recover that

$$0 = s_1 ds_2 + s_2 ds_1,$$

and consequently $ds_2 = -s_2^2 ds_1$ via $s_1^{-1} = s_2$. So $\Omega^1_{\mathbb{G}'_{\mathrm{m}}|\mathbb{G}_{\mathrm{m}}}(D_f)$ is generated only by ds_1. But the relation $s_1^n = x \in A$ tells us that $n s_1^{n-1} ds_1 = 0$, and since $n s_1^{n-1} \neq 0$ is invertible in B, we must have $ds_1 = 0$, hence $\Omega^1_{\mathbb{G}'_{\mathrm{m}}|\mathbb{G}_{\mathrm{m}}}(D_f) = 0$. It follows from this that $\Omega^1_{\mathbb{G}'_{\mathrm{m}}|\mathbb{G}_{\mathrm{m}}}$ is zero on every open of $\mathbb{G}_{\mathrm{m}}$, in such a case we write $\Omega^1_{\mathbb{G}'_{\mathrm{m}}|\mathbb{G}_{\mathrm{m}}} = 0$;

(ii) Recall the affine scheme $X := \mathbb{A}^2_{\mathbb{C}} \setminus \{0\}$ with affine charts $U_1 := \mathrm{Spec}(\mathbb{C}[x, y, x^{-1}])$ and $U_2 := \mathrm{Spec}(\mathbb{C}[x, y, y^{-1}])$. The inclusion of $\mathbb{C}$ into the coordinate ring gives us a canonical projection onto $C := \mathrm{Spec}(\mathbb{C})$. Relative differentials $\Omega^1_{X|C}(U_1)$ are generated by $\{dx, dy\}$ over $\mathbb{C}[x, y, x^{-1}]$ (we see $d(x^{-1}) = -x^{-2}dx$), and relative differentials $\Omega^1_{X|C}(U_2)$ are again generated by $\{dx, dy\}$, though this time over $\mathbb{C}[x, y, y^{-1}]$. One then computes that $\Omega^1_{X|C}(X) = \Omega^1_{X|C}(U_1 \cup U_2)$ is generated by $\{dx, dy\}$ over $\mathbb{C}[x, y]$. Heuristically, we may see $\Omega^1_{X|C}(X)$ as the set of differentials on $\Omega^1_{X|C}(U_1)$ which are defined on U_2, or vice versa;

(iii) Suppose that L/K is an extension of number fields, that it is unramified away from a finite set S of prime ideals of $\mathcal{O}_K$, and let $Y := \mathrm{Spec}(\mathcal{O}_L)$ and $X := \mathrm{Spec}(\mathcal{O}_K)$. Consider then the inclusion map $\pi^\# : \mathcal{O}_K \hookrightarrow \mathcal{O}_L$, which induces an epimorphism $\pi : Y \twoheadrightarrow X$ of affine schemes. We then wish to compute $\Omega^1_{Y|X}$ on, say, the basic open $D_f \subseteq X$. Computing these relative differentials then comes down to computing the relative differentials of the map $\pi^\# : \mathcal{O}_{K,f} \hookrightarrow \mathcal{O}_{L,f}$. Since L is a finite dimension K-vector space, it follows that $\mathcal{O}_L$ is a finitely generated $\mathcal{O}_K$-module, and so suppose we have some generators $\mathcal{O}_L = \mathcal{O}_K[\alpha_1, \ldots, \alpha_n]$, and consequently $\mathcal{O}_{L,f} = \mathcal{O}_{K,f}[\alpha_1, \ldots, \alpha_n]$. We know then that $\Omega^1_{Y|X}(D_f) = \Omega^1_{\mathcal{O}_{L,f}|\mathcal{O}_{K,f}}$ must be generated by $\{d\alpha_1, \ldots, d\alpha_n\}$ as an $\mathcal{O}_{L,f}$-module. Since each α_i is algebraic over K, and in particular lies in the integral closure of $\mathcal{O}_K$, we may consider the minimal polynomial $m_i \in \mathcal{O}_K[t]$ of each α_i. The relation $m_i(\alpha_i) = 0$ gives

the relation $m_i'(\alpha_i)d\alpha_i = 0$ by Proposition 2.23. Such $m_i'(\alpha_i)$ is invertible in $\mathcal{O}_L$ exactly when $\mathcal{O}_K[\alpha_i]$ is an everywhere-unramified extension[3] of $\mathcal{O}_K$, so $d\alpha_i = 0$ exactly when α_i generates an unramified extension. If $\{\alpha_1, \dots, \alpha_s\}$ are the subset of generators which generate ramified extensions, then $\Omega^1_{Y|X}(D_f)$ is generated by $\{d\alpha_1, \dots, d\alpha_s\}$. In particular, L/K is everywhere-unramified if and only if $\Omega^1_{Y|X} = 0$;

(iv) With exactly the same notation as (iii), we let $X' := \mathrm{Spec}(\mathcal{O}_{K,S})$ and $Y' = \mathrm{Spec}(\mathcal{O}_{L,S})$. Then since each $m_i'(\alpha_i)$ is invertible in $\mathcal{O}_{K,S}$, we see that $\Omega^1_{Y'|X'} = 0$.

♢

2.4.3 Finite étale morphisms, and their relations to smooth and unramified morphisms

Recall the notion of the *residue field* $k(\mathfrak{p})$ of a scheme S at a point $\mathfrak{p} \in S$, which is defined the be the residue of the local ring $\mathcal{O}_{S,\mathfrak{p}}$.

Definition 2.26. Let $f : X \to S$ be a morphism of schemes which is locally of finite type. We say that f is *finite étale* (or *fét* for short) if the fibre $X_\mathfrak{p} := X \times_S \mathrm{Spec}(k(p))$ of f at a point $\mathfrak{p} \in S$ is the spectrum of a finite étale $k(\mathfrak{p})$-algebra (i.e. a finite sum of finite separable extensions of $k(p)$). In particular, a morphism $f : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ of affine schemes which is locally of finite type is finite étale if and only if $B \otimes_A k(\mathfrak{p})$ is a finite étale $k(\mathfrak{p})$-algebra for every $\mathfrak{p} \in S$. When a surjective morphism $f : X \twoheadrightarrow S$ is finite étale, we may refer to it as a *finite étale cover*.

Example 2.27. (i) Consider the morphism $\mathbb{G}'_\mathrm{m} \twoheadrightarrow \mathbb{G}_\mathrm{m}$; $z \mapsto z^n$ from Example 2.25(i). The residue field at $\mathfrak{p} \in \mathbb{G}_\mathrm{m}$ is exactly the residue field of the local ring $\mathbb{C}[x, x^{-1}]_\mathfrak{p}$, i.e. it is given by $k(\mathfrak{p}) = \mathrm{Frac}(\mathbb{C}[x, x^{-1}]/\mathfrak{p})$. The points of $\mathbb{G}_\mathrm{m} = \mathbb{G}'_\mathrm{m}$ are either closed or generic, since $\mathbb{G}_\mathrm{m}$ is dimension 1 (this is just the Krull dimension for affine schemes), so at the generic point $\mathfrak{p} = (0) \in \mathbb{G}'_\mathrm{m}$ we see that the fibre is the spectrum of $\mathbb{C}[x^{1/n}, x^{1/n}] \otimes_{\mathbb{C}[x,x^{-1}]} \mathbb{C}(x) = \mathbb{C}(x^{1/n})$, which is a separable extension of $\mathbb{C}(x)$. Closed points $\mathfrak{m} \in \mathbb{G}_\mathrm{m}$ are just maximal ideals of $\mathbb{C}[x, x^{-1}]$, which are all of the form $\mathfrak{m}_a := (x - a, x^{-1} - a^{-1}) = (x - a)$ for some $a \in \mathbb{C}^\times$, and we see that

$$k(\mathfrak{m}_a) = \frac{\mathbb{C}[x, x^{-1}]}{\mathfrak{m}_a} = \frac{\mathbb{C}[x, x^{-1}]}{(x - a)} \cong \mathbb{C}.$$

A closed point $\mathfrak{n}_b := (t - b)$ of $\mathbb{C}[t, t^{-1}] \cong \mathbb{C}[x^{1/n}, x^{-1/n}]$ maps to $\mathfrak{m}_a$ under this morphism if and only if $b^n = a$, so there are n distinct points mapping to $\mathfrak{m}_a$. So, we obtain that the fibre at $\mathfrak{m}_a$ is the spectrum of

$$\mathbb{C}[t, t^{-1}] \otimes_{\mathbb{C}[x,x^{-1}]} \mathbb{C} \cong \frac{\mathbb{C}[t, t^{-1}]}{(t - \sqrt[n]{a})} \oplus \frac{\mathbb{C}[t, t^{-1}]}{(t - \zeta_n \sqrt[n]{a})} \oplus \cdots \oplus \frac{\mathbb{C}[t, t^{-1}]}{(t - \zeta_n^{n-1} \sqrt[n]{a})} \cong \mathbb{C}^{\oplus n},$$

where $\zeta_n = e^{2\pi i/n}$. Since $\mathbb{C}/\mathbb{C}$ is trivially a finite separable extension, $\mathbb{C}^{\oplus n}$ is a finite étale $\mathbb{C}$-algebra and thus the morphism $\mathbb{G}'_\mathrm{m} \twoheadrightarrow \mathbb{G}_\mathrm{m}$; $z \mapsto z^n$ is finite étale. We will see later that this is the equivalent of the n-fold cover of the circle $S^1 \twoheadrightarrow S^1$ from the classical theory of covering spaces;

(ii) If L/K is a finite extension of number fields, unramified away from a finite set S of primes of $\mathcal{O}_K$. Then the map $\pi : \mathrm{Spec}(\mathcal{O}_{L,S}) \twoheadrightarrow \mathrm{Spec}(\mathcal{O}_{K,S})$ induced by inclusion of S-integers $\pi^\# : \mathcal{O}_{K,S} \hookrightarrow \mathcal{O}_{L,S}$ is finite étale. We'll see this in more detail in §2.4.5.

♢

While this definition is nice and computable, we lose the benefit of geometric intuition. We will define some more geometric properties which will turn out to give an equivalent definition.

Definition 2.28. Let $f : X \to S$ be a morphism of schemes which is locally of finite presentation. Then we say that f is *unramified* if $\Omega^1_{X|S} = 0$ with respect to f. We say that f is *smooth at* $x \in X$ there exists affine charts $U = \mathrm{Spec}(B) \ni x$ and $V = \mathrm{Spec}(A) \ni f(x)$, with $B = A[t_1, \dots, t_n]/(p_1, \dots, p_m)$, such that the B-ideal generated by the $m \times m$ minors of the matrix $(\partial p_i/\partial t_j)_{ij}$ is B. We say that f is *smooth* if it is smooth at each $x \in X$.

[3] This is because $m_i'(\alpha_i)$ generates the different $\mathfrak{D} := \mathfrak{D}_{K(\alpha_i)/K} \trianglelefteq \mathcal{O}_K[\alpha_i]$ (cf. [FC67, Prop. I.4.6(i)]) which is divisible by a prime $\mathfrak{P}$ of $\mathcal{O}_K[\alpha_i]$ if and only if $\mathfrak{P}$ ramifies. Therefore, $K(\alpha_i)/K$ is unramified if and only if $\mathfrak{D} := (m_i'(\alpha_i))$ is not divisible by any prime ideals of $\mathcal{O}_K[\alpha_i]$, i.e. if and only if $\mathfrak{D} = (1)$, i.e. if and only if $m_i'(\alpha_i)$ is invertible.

Together these form an equivalent definition of 'finite étale', but we omit the proof of this because it really isn't all too important for us.

Theorem 2.29. *Let $f : X \to S$ be a morphism of schemes which is locally of finite presentation. Then f is finite étale if and only if f is both smooth and unramified.*

Proof. Omitted; see [Gro67, Corollaire 17.6.2]. □

2.4.4 ...for complex varieties

These definitions are all a bit abstract, so to give some intuition we'll see what this means for a morphism $f : Y \to X$ of affine complex varieties (such a map is automatically locally finitely presented). In this case, the closed points of each of these can be embedded into some $\mathbb{C}^n$ and $\mathbb{C}^m$ respectively, and thus can be endowed with the Euclidean topology to be made into complex manifolds. The corresponding complex manifolds will be denoted by X^{an} and Y^{an}.

Since morphisms of varieties are polynomial maps, the map $f : Y \to X$ induces a map $f : Y^{\text{an}} \to X^{\text{an}}$ of manifolds. Let $q \in Y^{\text{an}}$ be a point and $p = f(q)$, both considered closed subvarieties of Y and X respectively which we may identify with $\text{Spec}(\mathbb{C})$. We will look at what both the 'unramified' and 'smooth' condition mean for this map $f : Y^{\text{an}} \to X^{\text{an}}$.

Unramified: Since dualising vector spaces $V \mapsto V^\vee$ is an involution, we get a series of identifications

$$\Omega^1_{X|p} = \Omega^1_{\mathscr{O}(X)|\mathbb{C}} = \text{Hom}_{\mathbb{C}}(\Omega^1_{\mathscr{O}(X)|\mathbb{C}}, \mathbb{C})^\vee = \text{Der}_{\mathbb{C}}(\mathscr{O}(X), \mathbb{C})^\vee = T_p^* X^{\text{an}}$$

where $T_p^* X^{\text{an}}$ denotes the cotangent space of X^{an} at p, and similarly $\Omega^1_{Y|q} = T_q^* Y^{\text{an}}$. Then the canonical exact sequence induced by the map[4] $\Omega^1_{X|p} \to \Omega^1_{Y|p}$ is of the form:

$$\underbrace{\Omega^1_{X|p}}_{T_p^* X^{\text{an}}} \longrightarrow \underbrace{\Omega^1_{Y|q}}_{T_q^* Y^{\text{an}}} \longrightarrow \Omega^1_{Y|X} \longrightarrow 0,$$

for which the first arrow may be identified with the adjoint $(d_q f)^\vee$ of the derivative $d_q f : T_q Y^{\text{an}} \to T_p X^{\text{an}}$ of f at q. So we see that f is unramified, i.e. $\Omega^1_{Y|X} = 0$, if and only if $(d_q f)^\vee$ is surjective. That is to say, f is unramified if and only if the derivative $d_q f$ is *injective*. The point here is that when unramified morphisms don't fold away dimensions.

Smooth: This one does not just translate to 'f is a smooth map', but rather that f is a smooth *submersion* (by the submersion theorem). That is, f is algebraically smooth exactly when the derivative $d_q f : T_q Y^{\text{an}} \to T_p X^{\text{an}}$ is *surjective*.

Together, these give us a nice characterisation of finite étale morphisms of affine varieties over $\mathbb{C}$:

Theorem 2.30. *A morphism $f : Y \to X$ of affine complex varieties is finite étale if and only if the corresponding map $f : Y^{an} \to X^{an}$ of complex manifolds is a local diffeomorphism.*

This should give us some nice intuition: we should think of a general finite étale morphism as the algebraic equivalent of a local diffeomorphism!

2.4.5 ...and for number rings

Another example of interest of us is extensions of number fields, and their assoicated S-integers. Once again, in this example the 'finite étale' condition is easy to characterise, although this time it is in terms of arithmetic data rather than geometric data.

Let L/K be a finite extension of number fields, and let S be a set of prime ideals of $\mathcal{O}_K$. We recall that the ring $\mathcal{O}_{K,S}$ of S-integers is formed from $\mathcal{O}_K$ by inverting generators of all of the prime ideals $\mathfrak{p} \in S$. When $\hat{S}$ is the set of primes of $\mathcal{O}_L$

[4] ∗ **Exc**. Look back at Example 2.22(i) to see if you can figure out what the map is yourself :)

above all of those in S, we may write $\mathcal{O}_{L,S} := \mathcal{O}_{L,\hat{S}}$ for the $\hat{S}$-integers of L. The inclusion $K \hookrightarrow L$ then restricts to an inclusion $\pi^{\#} : \mathcal{O}_{K,S} \hookrightarrow \mathcal{O}_{L,S}$ of S-integers, which in turn induces a morphism $\pi : \mathrm{Spec}(\mathcal{O}_{L,S}) \twoheadrightarrow \mathrm{Spec}(\mathcal{O}_{K,S})$ of affine schemes. Such a morphism is always locally finitely presented.

At the generic point $(0) \in \mathrm{Spec}(\mathcal{O}_{K,S})$, the residue field of $\mathrm{Spec}(\mathcal{O}_{K,S})$ is given by $\mathrm{Frac}(\mathcal{O}_{K,S}) = K$, and so the fibre of π at (0) is given by the spectrum of

$$\mathcal{O}_{L,S} \otimes_{\mathcal{O}_{K,S}} K = L,$$

which is always a finite étale K-algebra, since L/K is a separable extension. But at a closed point $\mathfrak{p} \in \mathrm{Spec}(\mathcal{O}_{K,S})$, things get more interesting. A nonzero prime ideal of $\mathcal{O}_{K,S}$ may be identified with a prime ideal of $\mathcal{O}_K$ not contained in S, and so we take this convention. The fibre of π here is given by the spectrum of

$$\mathcal{O}_{L,S} \otimes_{\mathcal{O}_{K,S}} \left(\frac{\mathcal{O}_{K,S}}{\mathfrak{p}} \right) = \frac{\mathcal{O}_{L,S}}{\mathfrak{p}\mathcal{O}_{L,S}} = \frac{\mathcal{O}_L}{\mathfrak{p}\mathcal{O}_L}.$$

Let $\mathfrak{P}_1, \ldots, \mathfrak{P}_r$ be the primes of $\mathcal{O}_L$ above $\mathfrak{p}$ (each pairwise distinct) and $e_1, \ldots, e_r \in \mathbb{N}$ such that $\mathfrak{p}\mathcal{O}_L = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}$, then Sunzi's theorem tells us that

$$\frac{\mathcal{O}_L}{\mathfrak{p}\mathcal{O}_L} = \frac{\mathcal{O}_L}{\mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}} \cong \frac{\mathcal{O}_L}{\mathfrak{P}_1^{e_1}} \oplus \cdots \oplus \frac{\mathcal{O}_L}{\mathfrak{P}_r^{e_r}}. \tag{$*$}$$

In particular, when each $e_i = 1$ we see that this is a sum of finite separable extensions of $k(\mathfrak{p}) = \mathcal{O}_K/\mathfrak{p}$. If $e_{i_0} > 1$ for some i_0, we see that $\mathcal{O}_L/\mathfrak{P}_{i_0}^{e_{i_0}}$ has nilpotents, so $\mathcal{O}_L/\mathfrak{P}_{i_0}^{e_{i_0}}$ cannot be a finite étale over $k(\mathfrak{p})$. Consequently, neither can the RHS of $(*)$ be a finite étale algebra over $k(\mathfrak{p})$. So $\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L$ is finite étale over $k(\mathfrak{p})$ if and only if all of the $e_i = 1$, that is, if and only if $\mathfrak{p}$ is unramified in L/K. Thus...

Theorem 2.31. *Let L/K be a finite extension of number fields, and S a finite set of primes of $\mathcal{O}_K$. The morphism $\pi_S : \mathrm{Spec}(\mathcal{O}_{L,S}) \twoheadrightarrow \mathrm{Spec}(\mathcal{O}_{K,S})$ induced by inclusion of S-integers is finite étale if and only if S contains all of the primes ramified in L/K. In particular, $\pi_\varnothing : \mathrm{Spec}(\mathcal{O}_L) \twoheadrightarrow \mathrm{Spec}(\mathcal{O}_K)$ is finite étale if and only if L/K is everywhere unramified.*

3 Topological fundamental groups via covering spaces

3.1 The need for an algebraic definition of π_1

Recall that the usual fundamental group $\pi_1(X, x_0)$ is defined as the group of homotopy classes of loops in X based at $x_0 \in X$. Since any scheme has an underlying topological space, we may then ask why we shouldn't just define the fundamental group of a scheme to be the fundamental group of the underlying Zariski topological space. While we can do this, there are two main reasons why this definition is bad:

(i) The underlying topological space of a scheme is far from containing all of the information about said scheme. For instance, though $\mathrm{Spec}(\mathbb{F}_p)$ and $\mathrm{Spec}(\mathbb{C})$ are both one-point spaces (and thus homeomorphic), as schemes they are far from isomorphic.

(ii) Homotopy in the Zariski topology is far from meaningful. In fact, if $X := \mathrm{Spec}(R)$ is the spectrum of an integral domain R then the map $H : X \times [0,1] \to X$ given by $H(\mathfrak{p}, 0) = \mathfrak{p}$ and $H(\mathfrak{p}, s) = (0)$ for $s > 0$ defines an explicit deformation retract of X onto $\{(0)\}$, so the spectrum of integral domains is contractible!

In this section, we will realise a canonical isomorphism that allows for a definition of $\pi_1(X, x_0)$ which is completely independent of homotopy (although secretly it will work because of the existence of a universal covering space).

Warning. Since this section is mainly just for intuition, it may be a bit sparse on examples.

3.2 Covering spaces in more detail

All of our discussion of the classical theory of covering spaces will be done over 'nice' topological spaces (path connected and locally path connected). Recall from Definition 1.1 the notion of a *covering space*. These are continuous surjections that locally look like $X \sqcup \cdots \sqcup X \twoheadrightarrow X$, or more precisely, they are continuous surjections $p : \widehat{X} \twoheadrightarrow X$ such that X admits an open cover $\{U_i\}_i$ such that each $p^{-1}[U_i]$ is a disjoint union of copies of U_i.

Example 3.1. (i) Let $S^1 \subseteq \mathbb{C}$ denote the circle group. The map $p : S^1 \twoheadrightarrow S^1$ given by $z \mapsto z^n$ is both continuous and surjective by definition. We see that the we can cover S^1 by the two open subsets $U_\pm := S^1 \setminus \{\pm 1\}$, each homemorphic to an open interval. The preimage of $U_\pm$ looks like

$$p^{-1}[U_\pm] = S^1 \setminus \left\{ \pm e^{2\pi i k/n} \;\middle|\; k = 0, 1, \ldots, n-1 \right\},$$

and on each of the components of this p restricts to a homeomorphism. So this is indeed a covering space.

(ii) Similarly, let $\widetilde{p} : \mathbb{R} \twoheadrightarrow S^1$; $x \mapsto e^{2\pi i x}$. Then the preimage of each $U_\pm$ under this is

$$\widetilde{p}^{-1}[U_+] = \coprod_{n \in \mathbb{Z}} \left(n - \frac{1}{2}, n + \frac{1}{2} \right), \qquad \widetilde{p}^{-1}[U_-] = \coprod_{n \in \mathbb{Z}} (n, n+1),$$

and $\widetilde{p} : \mathbb{R} \twoheadrightarrow S^1$ restricts to a homeomorphism on each of the components of this. We see that $\widetilde{p}$ is also a covering space. ♢

Definition 3.2. Let X be a space, and $p_i : \widehat{X}_i \twoheadrightarrow X$ be two covering spaces ($i = 1, 2$). A *morphism of covering spaces* $\varphi : p_1 \to p_2$ is a continuous map $\varphi : X_1 \to X_2$ such that $p_1 = p_2 \circ \varphi$. The *category of covers of X*, $\mathbf{Cov}(X)$, is the category whose objects are covers of X, and whose morphisms are these morphisms of covers of X.

We note that this category comes equipped with a family of functors to **Set**, called the *fibre functors*. For $x_0 \in X$, the fibre functor at x_0 is given by

$$\mathrm{Fib}_{x_0} : \mathbf{Cov}(X) \to \mathbf{Set}; \quad p \mapsto p^{-1}[\{x_0\}].$$

The point of this section will be to prove the claim that the automorphism group of Fib_{x_0} is canonically isomorphic to the fundamental group $\pi_1(X, x_0)$, and thus provide us with a path for defining fundamental groups for schemes.

A lot of this will follow from the compatibility of covering spaces with lifts. There are two main results here: the *homotopy lifting property*, and the *unique lift property*.

Theorem 3.3 (Homotopy lifting property; HLP)**.** *Let X and Y be PC, LPC spaces. Let $p : \widehat{X} \twoheadrightarrow X$ be a covering space, $\{f_t : Y \to X\}_{t\in[0,1]}$ a homotopy of maps $Y \to X$, and $g_0 : Y \to \widehat{X}$ a continuous lift of f_0 through p. Then there exists a unique homotopy $\{\widehat{f}_t : Y \to \widehat{X}\}_{t\in[0,1]}$ lifting $\{f_t\}_t$ such that $\widehat{f}_0 = g_0$.*

Proof. Omitted. See [Hat01, Thm. 1.7] for a proof for $p : \mathbb{R} \to S^1$ which generalises to all covers. □

This has some immediate corollaries:

Corollary 3.3.1. Let X be a PC, LPC space, $p : \widehat{X} \twoheadrightarrow X$ a covering space, and $x_0 \in X$. Let $\gamma : [0,1] \to X$ be a path starting at x_0. For each $\widehat{x}_i \in p^{-1}[\{x_0\}]$ there is a unique path $\widehat{\gamma}_i : [0,1] \to \widehat{X}$ lifting γ starting at $\widehat{x}_i$.

Proof. This is immediate from the homotopy lifting property (Theorem 3.3) applied with $Y = \{*\}$. □

Corollary 3.3.2. Let $p : \widehat{X} \twoheadrightarrow X$ be a covering space, $x_0 \in X$, and $\widehat{x}_0 \in \widehat{X}$. Then p induces an injective homomorphism $p_* : \pi_1(\widehat{X}, \widehat{x}_0) \hookrightarrow \pi_1(X, x_0)$, and thus we may view $\pi_1(\widehat{X}, \widehat{x}_0)$ as a subgroup of $\pi_1(X, x_0)$, consisting of homotopy classes of the x_0-loops in X whose unique lift to $\widehat{X}$ at $\widehat{x}_0$ are also loops.

Proof. The homomorphism p_* is given by $p_*([\gamma]) := [p \circ \gamma]$, which we may verify to respect concatenation of paths. Injectivity follows from the homotopy lifting property (Theorem 3.3), since there is a *unique* lift of $p \circ \gamma$ based at $\widehat{x}_0$. □

Theorem 3.4 (Unique lift property; ULP)**.** *Let X and Y be PC, LPC spaces, $p : \widehat{X} \twoheadrightarrow X$ a covering space, and $f : Y \to X$ a continuous map. If $\widehat{f}_i : Y \to \widehat{X}$ are lifts of f through p ($i = 1, 2$), then they either agree everywhere or they agree nowhere. That is, if $\widehat{f}_1(y) = \widehat{f}_2(y)$ for* some *$y \in Y$, then $\widehat{f}_1 = \widehat{f}_2$.*

Proof. Let $A \subseteq Y$ be the set of $y \in Y$ such that $\widehat{f}_1(y) = \widehat{f}_2(y)$, and $B = Y \setminus A$ its complement. Our aim is to show that A and B are both open. Since Y is connected and $A \sqcup B = Y$, we will then be able to conclude that one of A or B has to be empty, and the other must be all of Y. If $y \in A$, then consider a neighbourhood $U \ni f(y)$ such that $p^{-1}[U] = \coprod_i \widehat{U}_i$ and p restricts to homeomorphisms $g_i := p|_{\widehat{U}_i} : \widehat{U}_i \xrightarrow{\sim} U$ is a homeomorphism (which exists since p is a cover). Let $V = f^{-1}[U]$, so that $f|_V : V \to U$. Then there must be a unique j such that $\widehat{f}_1(y) = \widehat{f}_2(y) \in \widehat{U}_j$, and since g_j is a homeomorphism such that $g_j \circ \widehat{f}_1|_V = f|_V = g_j \circ \widehat{f}_2|_V$, we conclude that

$$\widehat{f}_1|_V = g_j^{-1} \circ f|_V = \widehat{f}_2|_V : V \to \widehat{U}_j,$$

and consequently that if $y \in A$, then A contains some open neighbourhood $V \ni y$, i.e. A is open. We may similarly conclude, for $y' \in B$ and $U' \ni f(y')$ an open neighbourhood, that $p^{-1}[U'] = \coprod_i \widehat{U}'_i$ with similar homeomorphisms $g'_i = p|_{\widehat{U}'_i}$. Let $V' = f^{-1}[U']$, so that $f|_{V'} : V' \to U'$. Since $\widehat{f}_1(y') \neq \widehat{f}_2(y')$, we may also conclude that

$$\widehat{f}_1|_{V'} = g_{j_1}^{-1} \circ f|_{V'} : V' \to \widehat{U}'_{j_1} \quad \text{and} \quad \widehat{f}_2|_{V'} = g_{j_2}^{-1} \circ f|_{V'} : V' \to \widehat{U}'_{j_2},$$

which agree nowhere since their codomains $\widehat{U}'_{j_1}, \widehat{U}'_{j_2}$ are disjoint. So B contains V', hence is also open. Since Y is connected and $Y = A \sqcup B$, we thus conclude one of A, B is $\varnothing$, and the other is Y. □

Corollary 3.4.1 (Lifting criterion)**.** Let X and Y be PC, LPC spaces, $p : \widehat{X} \twoheadrightarrow X$ a covering space, and $f : Y \to X$. Let $y_0 \in Y$ and $f(y_0) = x_0$, and let $\widehat{x}_0 \in p^{-1}[\{x_0\}]$. Then there exists a lift $\widehat{f} : Y \to \widehat{X}$ with $\widehat{f}(y_0) = \widehat{x}_0$ if and only if $f_*\pi_1(Y, y_0) \subseteq p_*\pi_1(\widehat{X}, \widehat{x}_0)$. Such a lift is unique.

Proof. The 'only if' direction is pretty immediate: if such $\widehat{f} : Y \to \widehat{X}$ exists, then

$$f_*\pi_1(Y, y_0) = (p \circ \widehat{f})_*\pi_1(Y, y_0) = p_*(\widehat{f}_*\pi_1(Y, y_0)) \subseteq p_*\pi_1(\widehat{X}, \widehat{x}_0).$$

For the 'if' direction, for each $y \in Y$ choose some $\gamma_y : [0,1] \to Y$ joining y_0 to y. Then by Corollary 3.3.1 there is a unique lift $\widehat{\delta}_y : [0,1] \to \widehat{X}$ of $\delta_y := f \circ \gamma_y$ with $\widehat{\delta}_y(0) = \widehat{x}_0$. We define $\widehat{f}(y) := \widehat{\delta}_y(1)$, which may be verified to be a lift of f. The ULP (Theorem 3.4) gives us uniqueness. □

3.3 The universal covering space and deck transformations

For each $x_0 \in X$, we will construct a *universal* covering space $\widetilde{p} : \widetilde{X}_{x_0} \twoheadrightarrow X$, i.e. a covering space such that every other cover $p : \widehat{X} \twoheadrightarrow X$ is of the form $p = f \circ \widetilde{p}$, for $f \in \mathrm{Hom}_X(\widetilde{X}, \widehat{X})$ a surjective morphism of covers. This will turn out to be a representing object for our functor Fib_{x_0}, and will be the connection between $\pi_1(X, x_0)$ and $\mathrm{Aut}(\mathrm{Fib}_{x_0})$. To construct such a space, we will need another assumption on X – that it is *semilocally simply connected*.

Definition 3.5. Let X be a PC, LPC space. A neighbourhood $U \subseteq X$ is called *semi-simply-connected* (or *SSC*) if every loop in U is homotopic to a point in X, that is, if the induced map $\pi_1(U, u_0) \to \pi_1(X, u_0)$ is the zero map. We say that X is *semilocally simply connected* or (*SLSC*) if it admits a cover by SSC neighbourhoods.

We first define the set

$$\widetilde{X}_{x_0} := \frac{\{\alpha : [0,1] \to X \mid \alpha(0) = x_0\}}{\text{homotopy w/ fixed ends}}$$

of homotopy classes (with fixed endpoints) of paths in X starting at x_0. We will endow $\widetilde{X}_{x_0}$ with a topology by declaring basic open neighbourhoods of $[\gamma] \in \widetilde{X}_{x_0}$ to be of the form

$$\widetilde{U}_{[\gamma]} := \{[\gamma \cdot \eta] \mid \eta : [0,1] \to U \text{ has } \eta(0) = \gamma(1)\}$$

for $U \ni \gamma(1)$ an SSC neighbourhood. $*$ **Exc**. Check that this defines a basis for a topology.

Theorem 3.6 (Universality of $\widetilde{X}_{x_0}$)**.** *The space $\widetilde{X}_{x_0}$ is simply connected, and the map $\widetilde{p} : \widetilde{X}_{x_0} \twoheadrightarrow X$ given by $\widetilde{p}([\gamma]) := \gamma(1)$ is a covering space. Moreover, $\widetilde{X}_{x_0}$ carries a monodromy left-action of $\pi_1(X, x_0)$, and if $p : \widehat{X} \twoheadrightarrow X$ is another cover and $\widehat{x}_0 \in p^{-1}[\{x_0\}]$, then $\widehat{X}$ is canonically isomorphic to the orbit space $p_*\pi_1(\widehat{X}, \widehat{x}_0)\backslash\widetilde{X}_{x_0}$.*

Proof sketch. First we show that $\widetilde{X}_{x_0}$ is path connected by defining a canonical lift $\widetilde{\gamma} : [0,1] \to \widetilde{X}_{x_0}$ of each $\gamma : [0,1] \to X$ by $\widetilde{\gamma}(t) := [\gamma_t]$, where $\gamma_t(s) := \gamma(s)$ for $s \le t$ and $\gamma_t(s) := \gamma(t)$ for $s \ge t$. $*$ **Exc**. Prove that the map $\widetilde{p}$ is continuous and surjective. We see that the preimage of an SSC neighbourhood $U \subseteq X$ under $\widetilde{p}$ is exactly

$$\widetilde{p}^{-1}[U] = \coprod_{[\alpha] \in \pi_1(X, x_0)} \widetilde{U}_{[\alpha \cdot \gamma]}$$

where $\gamma : [0,1] \to X$ is any path ending in U, and since U is path connected and SSC the restriction $\widetilde{p}|_{\widetilde{U}_{[\gamma]}} : \widetilde{U}_{[\gamma]} \to U$ is bijective, and thus by definition of the topology on $\widetilde{X}_{x_0}$ this must be a homeomorphism. So $\widetilde{p}$ is a cover. For simply connectedness we show that $\widetilde{p}_*\pi_1(\widetilde{X}_{x_0}, \widetilde{x}_0) = 0$, where $\widetilde{x}_0 \in \widetilde{X}_{x_0}$ is the constant path at x_0. By Corollary 3.3.2, we see that

$$\widetilde{p}_*\pi_1(\widetilde{X}_{x_0}, \widetilde{x}_0) = \{[\alpha] \in \pi_1(X, x_0) \mid \widetilde{\alpha} \text{ is a loop}\}.$$

But the statement that $\widetilde{\alpha}$ is a loop says that $\widetilde{\alpha}(1) = \widetilde{\alpha}(0)$, and thus $\alpha_1 := \alpha$ is homotopic to α_0 (which is the constant path at x_0). That is, $\widetilde{p}_*\pi_1(\widetilde{X}_{x_0}, \widetilde{x}_0)$ is trivial.

The monodromy left-action of $\pi_1(X, x_0)$ is exactly by concatenation $[\alpha] \cdot [\gamma] := [\alpha \cdot \gamma]$. For a fixed cover $p : \widehat{X} \twoheadrightarrow X$ with basepoints $\widehat{x}_0 \in \widehat{X}$ and $x_0 \in X$, let $H := p_*\pi_1(\widehat{X}, \widehat{x}_0)$. We may then define $p_H : H\backslash\widetilde{X}_{x_0} \to X$ by $H[\gamma] \mapsto \gamma(1)$. Then we can verify $(p_H)_*\pi_1(H\backslash\widetilde{X}_{x_0}, H\widetilde{x}_0) = H = p_*\pi_1(\widehat{X}, \widehat{x}_0)$, and thus the lifting criterion (Corollary 3.4.1) tells us that $(\widehat{X}, \widehat{x}_0)$ and $(H\backslash\widetilde{X}_{x_0}, H\widetilde{x}_0)$ are canonically isomorphic as based covers of (X, x_0). □

Corollary 3.6.1. The functor $\mathrm{Fib}_{x_0} : \mathbf{Cov}(X) \to \mathbf{Set}$ is represented by $\widetilde{X}_{x_0}$.

Proof. Let $q : \widehat{X} \twoheadrightarrow X$ be a cover, and $x_0 \in X$. A morphism $\widetilde{X}_{x_0} \to \widehat{X}$ of covers is exactly a lift of $\widetilde{p} : \widetilde{X}_{x_0} \twoheadrightarrow X$ through q, and thus by the unique lifting property (Theorem 3.4) they are uniquely determined by where they send $\widetilde{x}_0 \in \widetilde{X}_{x_0}$. But again, by the definition of a morphism such $\widetilde{x}_0$ must be mapped into $\mathrm{Fib}_{x_0}(q)$. Thus we get an injection $\mathrm{Hom}_{\mathbf{Cov}(X)}(\widetilde{X}_{x_0}, \widehat{X}) \hookrightarrow \mathrm{Fib}_{x_0}(q)$ by $\varphi \mapsto \varphi(\widetilde{x}_0)$. To prove this is surjective, it is enough to use the lifting criterion (Corollary 3.4.1): since $\widetilde{X}_{x_0}$ is simply connected the image of $\pi_1(\widetilde{X}_{x_0}, \widetilde{x}_0)$ is contained in $q_*\pi_1(\widehat{X}, \widehat{x}_0)$, and thus for each $\widehat{x}_i \in \mathrm{Fib}_{x_0}(q)$ there exists a unique morpism $\varphi \in \mathrm{Hom}_{\mathbf{Cov}(X)}(\widetilde{X}_{x_0}, \widehat{X})$ sending $\widetilde{x}_0$ to $\widehat{x}_i$. □

The automorphisms of $\widetilde{X} := \widetilde{X}_{x_0}$ as a cover of X are called *deck transformations*. The Yoneda lemma then tells us that the group $\mathrm{Aut}(\widetilde{p})$ of deck transformations may be identified with $\mathrm{Aut}(\mathrm{Fib}_{x_0})$ for free! But we see, as a corollary of the 'free monodromy action' stipulation of Theorem 3.6, that the group of deck transformations may *also* be identified with $\pi_1(X, x_0)$. So in the end, we get a canonical isomorphism...

Corollary 3.6.2. $\pi_1(X, x_0) \cong \mathrm{Aut}(\mathrm{Fib}_{x_0})$, when X is PC, LPC, SLSC.

Example 3.7. It's not too hard to see that the universal covering space of S^1 is indeed $\mathbb{R}$. Not only is $\pi_1(\mathbb{R}, 0)$ trivial, but paths in S^1 (based at $1 = e^0 \in S^1$) are all homotopic to one of the 'standard' paths $\gamma_A : [0,1] \to S^1;\ x \mapsto e^{2\pi i A x}$, where $A \in \mathbb{R}$. We can 'unravel' this path through $\widetilde{p} : \mathbb{R} \twoheadrightarrow S^1$ to reveal a linear path from 0 to $A \in \mathbb{R}$, which may be identified with the point $A \in \mathbb{R}$.

Since $\pi_1(S^1, 1) \cong \mathbb{Z}$, we expect the group of automorphisms of $\widetilde{p} : \mathbb{R} \twoheadrightarrow S^1$ to be isomorphic to $\mathbb{Z}$. To each $n \in \mathbb{Z}$, the shift map $\mathbb{R} \to \mathbb{R}; x \mapsto x + n$ defines such a deck transformation, so the isomorphism of Corollary 3.6.2 is rather explicit in this case. ♢

3.4 The classification of covers

Putting all of this together allows us to wrap up the classical theory of covering spaces in a nice little bow. Subgroups of $\pi_1(X, x_0)$ should now determine covering spaces and vice versa.

Theorem 3.8 (Galois theory for covering spaces)**.** *Let X be a PC, LPC, SLSC space, and $x_0 \in X$. The map sending a based cover $p : (\widehat{X}, \widehat{x}_0) \twoheadrightarrow (X, x_0)$ to the subgroup $p_*\pi_1(\widehat{X}, \widehat{x}_0) \subseteq \pi_1(X, x_0)$ is a bijection (up to isomorphisms of covers). That is, isomorphism classes of based covers of X are in bijection with subgroups of $\pi_1(X, x_0)$.*

Proof. The inverse map is given by sending $H \leq \pi_1(X, x_0)$ to the orbit space $(H\backslash\widetilde{X}_{x_0}, H\widetilde{x}_0)$ under the monodromy action of $\pi_1(X, x_0)$ on $(\widetilde{X}_{x_0}, \widetilde{x}_0)$. □

Such a classification explicitly depends on the existence of a universal cover $\widetilde{X}_{x_0}$, but this is generally of infinite degree (i.e. the fibres are generally infinite). We run into a bit of an issue translating to the situation for schemes: our finite étale morphisms are just that – finite! Perhaps the more useful setting for us for covers of finite degree.

3.4.1 The classification of finite covers

For the last bit of this section, we are going to see what this theory looks like for finite covers. We've yet to define these, but this definition is really not too hard and relies on one observation.

Lemma 3.9. If $p : \widehat{X} \twoheadrightarrow X$ is a covering space and $x_0, x_1 \in X$, then $\mathrm{Fib}_{x_0}(p)$ and $\mathrm{Fib}_{x_1}(p)$ are in bijection.

Proof. This is a bit overkill (usually this is one of the first results you prove about covering spaces), but the universal covers $\widetilde{X}_{x_0}$ and $\widetilde{X}_{x_1}$ are isomorphic as (unbased) covers of X, and thus we get bijections

$$\mathrm{Fib}_{x_0}(p) \approx \mathrm{Hom}_{\mathbf{Cov}(X)}(\widetilde{X}_{x_0}, X) \approx \mathrm{Hom}_{\mathbf{Cov}(X)}(\widetilde{X}_{x_1}, X) \approx \mathrm{Fib}_{x_1}(p).$$ □

Definition 3.10. Let $p : \widehat{X} \twoheadrightarrow X$ be a covering space and $x \in X$. The cardinality $\#\mathrm{Fib}_x(p)$ is called the *degree* of p, denoted $\deg(p)$. We say p is a *finite cover* of X if $\deg(p)$ is finite. We denote by $\mathbf{FinCov}(X)$ the full subcategory of finite covers of X.

The fibre functor thus restricts to a functor

$$\mathrm{FinFib}_{x_0} : \mathbf{FinCov}(X) \to \mathbf{FinSet}.$$

The distinction here is now this functor is no longer generally representable, since universal covers need not be of finite degree. But by universality, we may identify $\widetilde{X}$ as the inverse limit of *all* of the finite covers of X. Thus, though Fib_{x_0} may be representable, our restricted functor FinFib_{x_0} is only *pro*-representable by a *pro*-object $\mathfrak{X} := \varprojlim_\alpha (\widehat{X}_\alpha, p_\alpha)$. To reiterate, the

inverse limit of all of these finite covers exists in $\mathbf{Cov}(X)$, but need not generally exist in $\mathbf{FinCov}(X)$. In this case we think of the universal cover as a formal inverse limit (this is roughly what is meant by *pro-object*; see a later talk). Consequently, the automorphism group of FinFib_{x_0} (again by the Yoneda lemma) may be given by

$$\mathrm{Aut}(\mathrm{FinFib}_{x_0}) = \mathrm{Aut}_{\mathrm{pro-}\mathbf{Cov}(X)}(\mathfrak{X}) = \varprojlim_{\alpha} \mathrm{Aut}_{\mathbf{Cov}(X)}(\widehat{X}_\alpha) \cong \widehat{\pi_1}(X, x_0),$$

where $\widehat{\pi_1}(X, x_0)$ denotes the profinite completion of $\pi_1(X, x_0)$. We gain from this a very valuable insight:

The equivalent of the fundamental group for finite covers is $\widehat{\pi_1}(X, x_0)$.

It is for this reason we should expect that our étale fundamental groups will be profinite. We then get another correspondence for finite covers, which may be more

Lemma 3.11. Let $p : (\widehat{X}, \widehat{x}_0) \to (X, x_0)$ be a based cover, and let $H := p_* \pi_1(\widehat{X}, \widehat{x}_0)$. Then $\deg(p) = [\pi_1(X, x_0) : H]$.

Theorem 3.12. *Let X be a PC, LPC, SLSC space, and $x_0 \in X$. Then we have a bijection*

$$\left\{\begin{array}{c}\text{Finite, based} \\ \text{covers of } (X, x_0)\end{array}\right\} \quad \rightleftarrows \quad \left\{\begin{array}{c}\text{Open subgroups} \\ \text{of } \widehat{\pi_1}(X, x_0)\end{array}\right\}$$

by sending a finite based cover $p : (\widehat{X}, \widehat{x}_0) \twoheadrightarrow (X, x_0)$ to the image of $p_* \pi_1(\widehat{X}, \widehat{x}_0)$ in $\widehat{\pi_1}(X, x_0)$.

Proof. The open subgroups of $\widehat{\pi_1}(X, x_0)$ are exactly the image of the finite index subgroups of $\pi_1(X, x_0)$. We then use Theorem 3.8 along with Lemma 3.11. □

4 Geometric points and the étale fundamental group

In this section we will build to a definition of the étale fundamental group analogous to that of §3. Namely, for S a scheme, we will consider the category $\mathbf{Fét}_S$ of finite étale covers of S and associate to it a fibre functor; the étale fundamental group will then be the automorphism group of this fibre functor.

4.1 Geometric points as basepoints

Fix a scheme S. We define the category $\mathbf{Fét}_S$, whose objects are schemes X equipped with finite étale covers $\phi : X \twoheadrightarrow S$, and whose morphisms $f : (X,\phi) \to (X',\phi')$ are morphisms of schemes such that $\phi = \phi' \circ f$. We may sometimes write just $X \in \mathbf{Fét}_S$ or $\phi \in \mathbf{Fét}_S$.

In order to complete our construction, we will need a good notion of the *fibre* of $(X,\phi) \in \mathbf{Fét}_S$ at a 'point' of S. The correct choice here is the notion of a *geometric fibre*.

Definition 4.1. A *geometric point* of S is a morphism $\overline{s} : \mathrm{Spec}(\Omega) \to S$, where Ω is an algebraically closed field. The *geometric fibre* of $(X,\phi) \in \mathbf{Fét}_S$ at $\overline{s}$ is then the pullback $\mathrm{Fib}_{\overline{s}}(X) := X \times_S \mathrm{Spec}(\Omega)$, thought of as its underlying set.

Example 4.2. (i) Let $S := \mathrm{Spec}(\mathbb{C}[t,t^{-1}]) = \mathbb{A}^1_{\mathbb{C}} \setminus \{0\}$ and $X := \mathrm{Spec}(\mathbb{C}[x,x^{-1}]) = \mathbb{A}^1_{\mathbb{C}} \setminus \{0\}$, considered with the finite étale cover $\phi_n : X \to S$ given by $\phi_n(z) := z^n$ (which is induced by $\mathcal{O}(S) \hookrightarrow \mathcal{O}(X)$; $t \mapsto x^n$). Consider the geometric point $\overline{s} : \mathrm{Spec}(\mathbb{C}) \hookrightarrow S$ associated to the evaluation map $\mathrm{ev}_1 : \mathbb{C}[t,t^{-1}] \twoheadrightarrow \mathbb{C}$ at 1. Using Proposition 2.13, we may then compute that the geometric fibre $\mathrm{Fib}_{\overline{s}}(X)$ of ϕ_n is given by

$$X \times_S \mathrm{Spec}(\mathbb{C}) = \mathrm{Spec}(\mathbb{C}[x,x^{-1}] \otimes_{\mathbb{C}[t,t^{-1}]} \mathbb{C}) = \mathrm{Spec}(\mathbb{C}[x,x^{-1}] \otimes_{\mathbb{C}[x^n,x^{-n}]} \mathbb{C}).$$

We may decompose $\mathbb{C}[x,x^{-1}] = \bigoplus_{r=0}^{n-1} x^r \mathbb{C}[x^n,x^{-n}]$, and thus we see that $\mathbb{C}[x,x^{-1}] \otimes_{\mathbb{C}[x^n,x^{-n}]} \mathbb{C} \cong \mathbb{C}^{\oplus n}$. So the geometric fibre is $\mathrm{Fib}_{\overline{s}}(X) = \mathrm{Spec}(\mathbb{C}^{\oplus n})$, which is just n points. When viewed as a subscheme of X, these correspond to the nth roots of unity in $\mathbb{C}^\times$;

(ii) Let K be a perfect field, L/K a finite separable field extension of degree n, and $\alpha \in L$ a primitive element (i.e. such that $L = K(\alpha)$). If $X = \mathrm{Spec}(L)$ and $S = \mathrm{Spec}(K)$, then we have already seen that the associated morphism $\phi : X \twoheadrightarrow S$ is a finite étale cover. Fix an algebraic closure $\overline{K}/K$ and let $\overline{s} : \mathrm{Spec}(\overline{K}) \to \mathrm{Spec}(K)$ be the associated geometric point. Then we can compute that the fibre $\mathrm{Fib}_{\overline{s}}(X)$ is given by

$$X \times_S \mathrm{Spec}(\overline{K}) = \mathrm{Spec}(L \otimes_K \overline{K}).$$

By properties of simple extensions, we know that $\{1,\alpha,\dots,\alpha^{n-1}\}$ is a K-basis for L, and thus we may think of $L = K \oplus \alpha K \oplus \cdots \oplus \alpha^{n-1}K$, and consequently

$$L \otimes_K \overline{K} = (K \oplus \alpha K \oplus \cdots \oplus \alpha^{n-1}K) \otimes_K \overline{K} \cong \overline{K}^{\oplus n},$$

and thus $\mathrm{Fib}_{\overline{s}}(X) = \mathrm{Spec}(\overline{K}^{\oplus n})$ consists of n points, which correspond to the n Galois conjugates of α in $\overline{K}$.

♢

It may seem like we are losing information simply by forgetting the Ω-scheme structure on $X \times_S \mathrm{Spec}(\Omega)$, but this structure is completely recoverable.

Proposition 4.3. Let $\phi : X \twoheadrightarrow S$ be a finite étale cover, and $\overline{s} : \mathrm{Spec}(\Omega) \to S$ a geometric point. The pullback $X \times_S \mathrm{Spec}(\Omega)$ may be identified with $\mathrm{Spec}(\Omega^{\oplus n})$ as a scheme, for some n. Consequently, $X \times_S \mathrm{Spec}(\Omega)$ is discrete.

Proof. Omitted, see [Sza09]. □

Proposition 4.4. Let S be a scheme and $\overline{s} : \mathrm{Spec}(\Omega) \to S$ a geometric point. The assignment $\mathrm{Fib}_{\overline{s}} : \mathbf{Fét}_S \to \mathbf{Set}$ of geometric fibres at $\overline{s}$ defines a functor.

Proof. The pullback over S is a categorical product in the category $\mathbf{Sch}_S$ of schemes over S, and thus any morphism $f : (X,\phi) \to (X',\phi')$ in $\mathbf{Fét}_S \subseteq \mathbf{Sch}_S$ induces a unique morphism $X \times_S \mathrm{Spec}(\Omega) \to X' \times_S \mathrm{Spec}(\Omega)$ of Ω-schemes. Forgetting the Ω-scheme structure, this defines a canonical function $\mathrm{Fib}_{\overline{s}}(X) \to \mathrm{Fib}_{\overline{s}}(X')$. □

Proposition 4.5. Let S be a connected scheme. Given two geometric points $\overline{s} : \mathrm{Spec}(\Omega) \to S, \overline{s}' : \mathrm{Spec}(\Omega') \to S$, there is a (non-canonical) natural isomorphism $\lambda : \mathrm{Fib}_{\overline{s}} \xRightarrow{\sim} \mathrm{Fib}_{\overline{s}'}$.

4.2 The étale fundamental group and first properties

Now that we have a fibre functor, we can use our reinterpretation of the usual fundamental group (§3) to give our first definition of the *étale fundamental group*.

Definition 4.6. Let S be a scheme, and $\overline{s} : \mathrm{Spec}(\Omega) \to S$ a geometric point. The *étale fundamental group* of S with respect to $\overline{s}$, denoted $\pi_1^{\text{ét}}(S, \overline{s})$, is the automorphism group of the fibre functor $\mathrm{Fib}_{\overline{s}} : \mathbf{Fét}_S \to \mathbf{Set}$.

We will see later that this group carries a canonical topology (the *Krull* topology), but for now we can just view it as a group. We get

Proposition 4.7. Let S be a scheme, $\overline{s} : \mathrm{Spec}(\Omega) \to S$ a geometric point, and $X \in \mathbf{Fét}_S$. There is a canonical (continuous) action of $\pi_1^{\text{ét}}(S, \overline{s})$ on $\mathrm{Fib}_{\overline{s}}(X)$, and the image of the associated homomorphism $\pi_1^{\text{ét}}(S, \overline{s}) \to \mathrm{Aut}_{\mathbf{Set}}(\mathrm{Fib}_{\overline{s}}(X))$ is canonically isomorphic to the group of $\mathbf{Fét}_S$-automorphisms of X, hence $\mathrm{Aut}_{\mathbf{Fét}_S}(X)$ is a (continuous) quotient of $\pi_1^{\text{ét}}(S, \overline{s})$.

Notation. We may write $\mathrm{Aut}(X/S)$ or $\mathrm{Aut}_S(X)$ for the $\mathbf{Fét}_S$-automorphism group of X.

Proof. The first observation is essentially by definition of natural transformations: to each $\eta \in \mathrm{Aut}(\mathrm{Fib}_{\overline{s}})$, we may take the action of η on X to be by its X-component $\eta_X \in \mathrm{Aut}(\mathrm{Fib}_{\overline{s}}(X))$. The continuity of the action with respect to the Krull topology will be seen in the next talk (it will follow immediately from the fact that Fib_S is pro-representable). Let G denote the image of $\pi_1^{\text{ét}}(S, \overline{s})$ in $\mathrm{Aut}_{\mathbf{Set}}(\mathrm{Fib}_{\overline{s}}(X))$. It is clear that there is a homomorphism $\mathrm{Aut}(X/S) \to G$ by restricting each $\phi \in \mathrm{Aut}(X/S)$ to fibres. For the converse, it shall suffice to prove this in the case that $X = \mathrm{Spec}(B)$ and $S = \mathrm{Spec}(A)$ are affine. In such a case, the geometric fibre of X is in bijection with the generators of B over A, and thus automorphisms of the fibre correspond to automorphisms of B as an A-algebra. The equivalence of categories $\mathbf{CAlg}_A^{\mathrm{op}} \simeq \mathbf{AffSch}_S$ thus tells us that each automorphism of the geometric fibre of X gives us a $\mathbf{Fét}_S$-automorphism of X. □

As a consequence of this proposition, we may now think of $\mathrm{Fib}_{\overline{s}}$ as a functor

$$\mathrm{Fib}_{\overline{s}} : \mathbf{Fét}_S \to \pi_1^{\text{ét}}(S, \overline{s}) - \mathbf{Set}$$

valued in the category of left-$\pi_1^{\text{ét}}(S, \overline{s})$-sets (one checks that each of the induced functions are $\pi_1^{\text{ét}}(S, \overline{s})$-equivariant).

Much as in the case of usual fundamental groups, changing basepoints corresponds to some conjugation isomorphism.

Proposition 4.8. Let S be a connected scheme, $\overline{s} : \mathrm{Spec}(\Omega) \to S$ and $\overline{s}' : \mathrm{Spec}(\Omega') \to S$ two geometric points, and $\lambda : \mathrm{Fib}_{\overline{s}} \xRightarrow{\sim} \mathrm{Fib}_{\overline{s}'}$ a natural isomorphism of fibre functors. Then λ induces an isomorphism of étale fundamental groups $\pi_1^{\text{ét}}(S, \overline{s}') \xrightarrow{\sim} \pi_1^{\text{ét}}(S, \overline{s})$ via $\phi \mapsto \lambda^{-1} \circ \phi \circ \lambda$. Consequently, the étale fundamental group of a connected scheme is independent of the basepoint, up to non-canonical isomorphism.

Proof. The latter observation follows from Proposition 4.5. For the former it suffices to note that, since λ is a natural isomorphism, we can do the same thing for $\lambda^{-1} : \mathrm{Fib}_{\overline{s}'} \xRightarrow{\sim} \mathrm{Fib}_{\overline{s}}$ to obtain an inverse homomorphism of étale fundamental groups. □

Corollary 4.8.1. If S is a connected scheme and $\pi_1^{\text{ét}}(S, \overline{s})$ is abelian for *some* geometric point $\overline{s}$, then $\pi_1^{\text{ét}}(S)$ is independent of the basepoint (up to canonical isomorphism).

Proof. Let $\overline{s}, \overline{s}'$ be two geometric points of S and $\lambda, \lambda' : \mathrm{Fib}_{\overline{s}} \xRightarrow{\sim} \mathrm{Fib}_{\overline{s}'}$ their associated homomorphisms. These induce isomorphisms $\Lambda, \Lambda' : \pi_1(S, \overline{s}') \xrightarrow{\sim} \pi_1(S, \overline{s})$, then $\Lambda \circ (\Lambda')^{-1}$ induces an inner automorphism of $\pi_1(S, \overline{s})$. If $\pi_1(S, \overline{s})$ is abelian, then any inner automorphism is the identity, and thus $\Lambda = \Lambda'$. Consequently, $\pi_1^{\text{ét}}(S, \overline{s})$ and $\pi_1^{\text{ét}}(S, \overline{s}')$ are canonically isomorphic for any choice of geometric points $\overline{s}, \overline{s}'$. □

4.2.1 Functoriality of the étale fundamental group

In much the same way that the topological π_1 is functorial, the algebraic $\pi_1^{\text{ét}}$ is functorial too.

Proposition 4.9. Let $\phi : S \to S'$ be a morphism of schemes, $\overline{s} : \mathrm{Spec}(\Omega) \to S$ a geometric point, and $\overline{s}' = \phi \circ \overline{s}$ the corresponding goemetric point. Then there exists a canonical homomorphism $\phi_* : \pi_1^{\text{ét}}(S, \overline{s}) \to \pi_1^{\text{ét}}(S', \overline{s}')$.

Proof. Consider the functor $\mathrm{BC}_{S',S} : \mathbf{Fét}_{S'} \to \mathbf{Fét}_S$ given by $\mathrm{BC}_{S',S}(X') := X' \times_{S'} S$. One then checks that $\mathrm{Fib}_{\overline{s}'} = \mathrm{Fib}_{\overline{s}} \circ \mathrm{BC}_{S',S}$. Thus, each automorphism of $\mathrm{Fib}_{\overline{s}}$ canonically induces an automorphism of $\mathrm{Fib}_{\overline{s}'}$ via this equality, and consequently we obtain a homomorphism $\phi_* : \pi_1^{\text{ét}}(S, \overline{s}) \to \pi_1^{\text{ét}}(S', \overline{s}')$. □

A A little bit about sheaves

The notion of a *sheaf* is extremely important to scheme theory, and allows us to talk about local properties in an organised way. If you've never seen anything about them, I'll try to introduce them as best as I can.

Example A.1. The main motivating example for a sheaf is the notion of sections of a continuous map $f : X \to Y$, that is, continuous maps $s : Y \to X$ such that $f \circ s = \mathrm{id}_X$. While not every continuous map admits such a continuous section, it is more likely that admit a *local* section, i.e. there is an open $U \subseteq X$ and some continuous $s : U \to Y$ such that $f \circ s = \mathrm{id}_U$. For each open set $U \subseteq X$, we will write $\mathcal{O}_f(U)$ for the set of local sections of f at U. Then we notice some important properties:

- **Restriction:** if $U \supseteq V$ and $s \in \mathcal{O}_f(U)$ is a local section at U, then the restriction $s|_V$ is a local section at V, i.e. $s|_V \in \mathcal{O}_f(V)$;
- **Locality:** if $U = \bigcup_i U_i$ and two sections $s, t \in \mathcal{O}_f(U)$ agree on each U_i (i.e. $s|_{U_i} = t|_{U_i}$ for each i), then they must agree on all of U, $s = t$. That is to say, sections are fully determined by their restrictions;
- **Gluing:** if $U = \bigcup_i U_i$ and for each i we have a section $s_i \in \mathcal{O}_f(U_i)$ which agree on overlaps (i.e. such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ whenever $U_i \cap U_j \neq \varnothing$). That is to say, we may glue together compatible sections.

This local-global structure is exactly the kind of thing that sheaves are meant to capture. ♢

Definition A.2. Let X be a topological space. A *presheaf of sets* (respectively, of abelian groups, of rings, etc.) on X is an assignment of a set (resp. abelian group, ring, etc.) $\mathcal{F}(U)$ to each open set $U \subseteq X$, along with a function (resp. homomorphism, morphism, etc.) $\rho^U_V : \mathcal{F}(U) \to \mathcal{F}(V)$ whenever $U \supseteq V$ which 'behave like restriction maps', i.e. $\rho^U_U = \mathrm{id}_U$ and $\rho^V_W \circ \rho^U_V = \rho^U_W$ for each tower $U \supseteq V \supseteq W$. An element $s \in \mathcal{F}(U)$ is called a *section* of $\mathcal{F}$ at U, and for $U = X$ a section $s \in \mathcal{F}(X)$ is called a *global section*. A presheaf $(\mathcal{F}, \rho)$ is said to be a *sheaf* if:

(L) for each open cover $U = \bigcup_i U_i$, two sections $s, t \in \mathcal{F}(U)$ are equal if and only if $\rho^U_{U_i}(s) = \rho^U_{U_i}(t)$ for each i; and

(G) for each open cover $U = \bigcup_i U_i$ and family of sections $\{s_i \in \mathcal{F}(U_i)\}_i$ such that

$$\rho^{U_i}_{U_i \cap U_j}(s_i) = \rho^{U_j}_{U_i \cap U_j}(s_j), \qquad U_i \cap U_j \neq \varnothing,$$

there is a section $s \in \mathcal{F}(U)$ such that $s_i = \rho^U_{U_i}(s)$.

A presheaf $\mathcal{F}$ satisfying property (L) is called *separated*.

As before, property (L) just says that sections are fully determined by their restrictions, and property (G) just says that we can glue sections together. If $\mathcal{F}$ is a presheaf whose local sections are functions, we see that (L) is automatically separated, since functions are fully determined by their values.

Example A.3. (i) As before, $\mathcal{O}_f$ defines a sheaf of sets on X;

(ii) The assignment $\mathcal{F} = \mathrm{Map}(-, \mathbb{R})$ on open sets of X, where $\mathrm{Map}(U, \mathbb{R})$ denotes the $\mathbb{R}$-algebra of continuous maps $U \to \mathbb{R}$, is a presheaf of $\mathbb{R}$-algebras with $\rho^U_V(f) := f|_V$. Definition of equality of functions tells us that $\mathcal{F}$ is separated, and the gluing lemma ensures that it satisfies property (G). Thus, $\mathcal{F}$ defines a sheaf on X;

(iii) **Non-example:** The assignment $\mathrm{Map}_{\mathrm{bdd}}(-, \mathbb{R})$ of bounded continuous functions still defines a separated presheaf with usual restrictions. However, we do not necessarily have property (G), since for example the function $\mathrm{id}_{\mathbb{R}} : \mathbb{R} \to \mathbb{R}$ is unbounded and continuous, but with respect to the open cover $\mathbb{R} = \bigcup_{n \in \mathbb{Z}} (n-1, n+1)$, its local sections are all bounded. These restrictions thus do not glue to a bounded function;

(iv) **Non-example:** We may define a separated presheaf $\underline{\mathbb{Z}}^{\mathrm{psh}}$ of commutative rings on any topological space X by assigning $\underline{\mathbb{Z}}^{\mathrm{psh}}(U) := \mathbb{Z}$ and $\rho^U_V = \mathrm{id}_{\mathbb{Z}}$. Any section of $\underline{\mathbb{Z}}^{\mathrm{psh}}$ is just an integer, so if we take two disjoint opens U, V, then the sections $1 \in \underline{\mathbb{Z}}^{\mathrm{psh}}(U), 2 \in \underline{\mathbb{Z}}^{\mathrm{psh}}(V)$ do not glue to a section on $U \cup V$. This is called the *constant presheaf* associated to $\mathbb{Z}$;

(v) We may instead define the presheaf $\underline{\mathbb{Z}}$ the assign to each U the set of locally constant functions $U \to \mathbb{Z}$. Then we find that we do, indeed, have properties (L) and (G), so this does define a sheaf of commutative rings on X. This is called the *constant sheaf* associated to $\mathbb{Z}$.

♢

For those that are more categorically minded, a presheaf of sets (resp. of abelian groups, etc.) on X is exactly a functor $\mathcal{F} : \mathrm{Open}(X)^{\mathrm{op}} \to \mathbf{Set}$ (resp. $\mathcal{F} : \mathrm{Open}(X)^{\mathrm{op}} \to \mathbf{Ab}$, etc.), where $\mathrm{Open}(X)$ is the preorder category of open sets of X under inclusion. Consequently, a sheaf of, say, commutative rings on X is a functor $\mathcal{F} : \mathrm{Open}(X)^{\mathrm{op}} \to \mathbf{CRing}$ such that for every open cover $U = \bigcup_{i \in I} U_i$ the sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F}(U) & \longrightarrow & \displaystyle\prod_{i \in I} \mathcal{F}(U_i) & \longrightarrow & \displaystyle\prod_{i,j \in I} \mathcal{F}(U_{i,j}) \\ & & s & \longmapsto & (s_i := s|_{U_i})_{i \in I} & \longmapsto & (s_i|_{U_{i,j}} - s_j|_{U_{i,j}})_{i,j} \end{array}$$

is exact, where the $\mathcal{F}(U) \to \mathcal{F}(V)$; $s \mapsto s|_V$ maps are those induced by $V \hookrightarrow U$, and $U_{i,j} := U_i \cap U_j$ for each $i, j \in I$.

As is common in analysis and topology, sometimes we want to study the behaviour of functions in an 'infinitesimal neighbourhood' of a point. In analysis and topology, this is achieved with limits. Similar we can use the categorical notion of a (co)limit to do this with sheaves.

Definition A.4. Let $\mathcal{F}$ be a sheaf on X. Then the *stalk* of $\mathcal{F}$ at $x \in X$, denoted $\mathcal{F}_x$, is the colimit $\mathcal{F}_x := \mathrm{colim}_{U \ni x} \mathcal{F}(U)$ with respect to the restriction morphisms. An element $g \in \mathcal{F}_x$ is called a *germ* of $\mathcal{F}$ at x.

Example A.5. Let $\mathcal{F} = \mathrm{Map}(-, \mathbb{R})$ be the sheaf of continuous maps to $\mathbb{R}$ on a topological space X (per Example A.3(i)). We may explicitly construct the stalk $\mathcal{F}_x$ at $x \in X$ as follows: we consider the set F_x of pairs (s, U) consisting of a section $s \in \mathcal{F}(U)$ on an open neighbourhood $U \ni x$. Then each $(s, U) \in F_x$ is definitionally contained in some $\mathcal{F}(U)$, so it should have some image in $\mathcal{F}_x$. Thus we get a map $F_x \to \mathcal{F}_x$ which must be surjective since $\mathcal{F}_x$ is a colimit. We will denote this map by $(s, U) \mapsto [s, U]$. We see that two germs $[s, U], [t, V] \in \mathcal{F}_x$ are equal if and only if the image of (s, U) and of (t, V) are equal in some $\mathcal{F}(W)$, again by the definition of colimits. That is, $[s, U] = [t, V]$ if and only if there is $W \subseteq U \cap V$ such that $s|_W = t|_W$. This recovers the classical notion of the *germ of a map*, hence the name *germ* for an element of a stalk. ♢

Example A.6. Let X be the closed subvariety of $\mathbb{A}^n$, say over $\mathbb{C}$. Then the *structure sheaf* $\mathcal{O}_X$ on X associates to a Zariski open U all of the *regular functions* $U \to \mathbb{A}^1$, that is, functions which are locally given by rational functions (one may think of this as the sheafification of the presheaf of rational functions). It is not immediately obivous that $\mathcal{O}_X(X)$ is equal to the coordinate ring $\mathcal{O}(X)$ of X, though this is true. On the basic open set $D(f) = p \in X \mid f(p) \neq 0$ for $f \in \mathcal{O}(X)$, we see that $\mathcal{O}_X(D(f)) = \mathcal{O}(X)_f$ is the localisation of $\mathcal{O}(X)$ at f, i.e. the ring resulting from adjoining an inverse of f, and restriction maps are simply the canonical inclusions as subrings of $\mathrm{Frac}\,\mathcal{O}(X)$. Consequently, we may think of colimits of these as unions, and thus the stalk at $p \in X$ may be computed as

$$\mathcal{O}_{X,p} = \operatorname*{colim}_{U \ni p} \mathcal{O}_X(U) = \bigcup_{U \ni p} \mathcal{O}_X(U) = \bigcup_{f \notin \mathfrak{m}_p} \mathcal{O}(X)_f = \mathcal{O}(X)_{\mathfrak{m}_p},$$

where $\mathfrak{m}_p$ is the maximal ideal of $\mathcal{O}(X)$ corresponding to p (see Section 2.1), and $\mathcal{O}(X)_{\mathfrak{m}_p}$ is the localisation at the maximal ideal $\mathfrak{m}_p$ (achieved by inverting all elements not in $\mathfrak{m}_p$). For instance when $X = \mathbb{A}^1$, then the stalk of $\mathcal{O}_X$ at $a \in \mathbb{A}^1 = \mathbb{C}$ is $\mathbb{C}[x]_{(x-a)}$, where we invert $(x - b)$ for every $b \neq a$. ♢

We also have a notion of a map between sheaves, called a *morphism*.

Definition A.7. Let $\mathcal{F}, \mathcal{G}$ be two sheaves of sets (resp. abelian groups, commutative rings, etc.) on X. A *morphism* $\varphi : \mathcal{F} \to \mathcal{G}$ consists of a function (resp. homomorphism, etc.) $\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$ commuting with restriction, i.e. such that for all opens $U \supseteq V$ the following diagram commutes:

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi_U} & \mathcal{G}(U) \\ \downarrow{\scriptstyle \rho^U_V} & & \downarrow{\scriptstyle \rho^U_V} \\ \mathcal{F}(V) & \xrightarrow{\varphi_V} & \mathcal{G}(V) \end{array}$$

When thinking of sheaves as functors from $\mathrm{Open}(X)^{\mathrm{op}}$, the notion of a morphism of sheaves coincides with the notion of a natural transformation between sheaves. The category of sheaves on X will be denoted $\mathbf{Sh}(X)$.

A.1 The étalé space and sheafification

Recall that our motivating example of a sheaf was the sheaf of sections of a continuous map $f : X \to Y$. This is a particularly important motivating example, since it turns out that every sheaf can be thought of in this way via a construction known as the *étalé space* of the sheaf. This space is usually a very strange topological space though, so these are not so easy to think about precisely.

If $\mathcal{F}$ is a sheaf on a space X, we have a canonical surjection

$$\pi : \coprod_{x \in X} \mathcal{F}_x \twoheadrightarrow X$$

where the stalk $\mathcal{F}_x$ is sent to $x \in X$. The idea of the étalé space construction is to endow $E(\mathcal{F}) := \coprod_x \mathcal{F}_x$ with a topology such that $\mathcal{F}$ naturally identifies with the sheaf of sections of π. Let $U \subseteq X$, then every $s \in \mathcal{F}(U)$ defines elements $s_x \in \mathcal{F}_x$ for each $x \in U$. We then define the subset $V(s, U) \subseteq E(\mathcal{F})$ to be the set containing all such germs of s, i.e.

$$V_{\mathcal{F}}(s, U) := \{s_x \in \mathcal{F}_x \mid x \in U\}.$$

Then such the collection of such sets covers $E(\mathcal{F})$ (i.e. $\bigcup_{s,U} V_{\mathcal{F}}(s, U) = E(\mathcal{F})$), and only $V_{\mathcal{F}}(s, U)$ and $V_{\mathcal{F}}(t, U')$ only intersect when U and U' intersect and $s|_{U''} = t|_{U''}$ for some $U'' \subseteq U \cap U'$; In such a case, we have

$$V_{\mathcal{F}}(s, U) \cap V_{\mathcal{F}}(t, U') = V_{\mathcal{F}}(s|_{U''}, U'').$$

Consequently, such sets form a basis of a topology on $E(\mathcal{F})$.

Definition A.8. The *étalé topology* on $E(\mathcal{F})$ is the topology $\mathcal{T}_{\text{ét}}$ generated by the sets $V_{\mathcal{F}}(s, U)$, where U ranges over all opens of X and s ranges over all of $\mathcal{F}(U)$. When $E(\mathcal{F})$ is equipped with its étalé topology, we call it the *étalé space* of $\mathcal{F}$.

Proposition A.9. The sheaf $\mathcal{O}_{\pi_{\mathcal{F}}}$ of continuous sections of $\pi_{\mathcal{F}} : E(\mathcal{F}) \twoheadrightarrow X$ canonically identifies with the sheaf $\mathcal{F}$.

Proof. Let $U \subseteq X$ be an open. Then a local section of π at U is exactly a function $s : U \to E(\mathcal{F})$ such that $\pi \circ s = \mathrm{id}_U$, that is, such that $s(u) \in \mathcal{F}_u$ for each $u \in U$. Such a section is continuous if and only if $s^{-1}[V_{\mathcal{F}}(t, U)] \subseteq X$ is open for each $U \subseteq X$ and $t \in \mathcal{F}(U)$. We see that

$$s^{-1}[V_{\mathcal{F}}(t, U)] = \{x \in X \mid s(x) \in V_{\mathcal{F}}(t, U)\} = \{x \in X \mid s(x) = t_x\}.$$

Any $t \in \mathcal{F}(U)$ naturally determines a unique $s_t \in \mathcal{O}_{\pi_{\mathcal{F}}}(U)$ by $s_t(x) := t_x$ (and if $\mathcal{F}$ is a sheaf of abelian groups, etc., this must be a homomorphism since taking germs is definitionally a homomorphism), so we get a canonical injective function (resp. homomorphism) $\mathcal{F}(U) \hookrightarrow \mathcal{O}_{\pi_{\mathcal{F}}}(U)$.

Now to prove surjectivity of this map. If $U' \subseteq U$, $s \in \mathcal{O}_{\pi_{\mathcal{F}}}(U)$, and $t \in \mathcal{F}(U')$, then $U'' := s^{-1}[V_{\mathcal{F}}(t, U')]$ is the largest open subset of U' on which $s(x) = t_x$. Taking an open cover $U = \bigcup_i U_i$, each U_i is contained in a preimage under s, and thus we get local sections $t_i \in \mathcal{F}(U_i)$ whose stalks agree with the values $s(x)$ on all of U_i. Since $\mathcal{F}$ is a sheaf, these glue together to a section $t \in \mathcal{F}(U)$. By construction, $s(x) = t_x$ for all $x \in U$. Thus, every $s \in \mathcal{O}_{\pi_{\mathcal{F}}}(U)$ is in the image of $\mathcal{F}(U)$, so this map is a canonical isomorphism. □

Corollary A.9.1. Every sheaf $\mathcal{F}$ on X is the sheaf of sections of its étalé space $E(\mathcal{F}) \twoheadrightarrow X$.

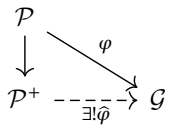
Corollary A.9.2. For every presheaf $\mathcal{P}$ on X, there exists a canonical sheaf $\mathcal{P}^+$ on X with a canonical morphism $\mathcal{P} \to \mathcal{P}^+$ such that $\mathcal{P}_x = \mathcal{P}^+_x$ for all $x \in X$. Moreover, if $\mathcal{F}$ is a sheaf then $\mathcal{F}^+ = \mathcal{F}$.

Proof. The construction of the étalé space and its corresponding sheaf of sections only requires $\mathcal{P}$ to be a presheaf. The map $\mathcal{P} \to \mathcal{P}^+$ is given by sending a section $t \in \mathcal{P}(U)$ to the section $t(x) := t_x$. Taking the sheaf of sections of $E(\mathcal{P}) \twoheadrightarrow X$ defines such a sheaf. The latter statement follows from Corollary A.9.1. □

Definition A.10. If $\mathcal{P}$ is a presheaf on X, then the sheaf $\mathcal{P}^+$ of Corollary A.9.2 is called the *sheafification* of $\mathcal{P}$.

Warning. Though the actual set $E(\mathcal{F})$ only depends on the stalks of $\mathcal{F}$, the étalé topology on $E(\mathcal{F})$ does depend on the sectional structure of $\mathcal{F}$, so a sheaf $\mathcal{F}$ is not necessarily fully determined by its stalks.

The sheafification $\mathcal{P}^+$ is the best possible approximation of $\mathcal{P}$ in the sense of the following universal property: if $\mathcal{G}$ is a sheaf on X and $\varphi : \mathcal{P} \to \mathcal{G}$ is a morphism of presheaves, then there exists a unique morphism of sheaves $\widehat{\varphi} : \mathcal{P}^+ \to \mathcal{G}$ making the diagram



commute.

A.2 New sheaves from old

In classical algebraic geometry, pullbacks and pushforwards allow us to pass maps from one variety to another. More generally, we should be able to do this for sections of a sheaf. In this section, we will take a continuous map $f : X \to Y$ and produce from it two functors

$$f_* : \mathbf{Sh}(X) \to \mathbf{Sh}(Y), \qquad f^* : \mathbf{Sh}(Y) \to \mathbf{Sh}(X),$$

called the *pushforward* and *pullback* functors.

Definition A.11. Let $f : X \to Y$ be a continuous map and $\mathcal{F}$ be a sheaf on X. We define the *pushforward* $f_*\mathcal{F}$ of $\mathcal{F}$ along f to be the presheaf on Y given by $f_*\mathcal{F}(U) := \mathcal{F}(f^{-1}[U])$.

Proposition A.12. If $f : X \to Y$ is continuous and $\mathcal{F}$ is a sheaf on X, then the pushforward $f_*\mathcal{F}$ presheaf defines a sheaf on Y. Moreover, such an assignment $f_* : \mathbf{Sh}(X) \to \mathbf{Sh}(Y)$ is functorial.

Proof. It suffices to note that the preimage of an open cover $U = \bigcup_{i\in I} U_i$ in Y under a continuous map $f : X \to Y$ gives an open cover $f^{-1}[U] = \bigcup_{i\in I} f^{-1}[U_i]$ in X. Thus, locality and gluing of $\mathcal{F}$ immediately imply that $f_*\mathcal{F}$ satisfies properties (L) and (G), thus defines a sheaf on Y. If $\varphi : \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves on X, then we may define a morphism $f_*\varphi : f_*\mathcal{F} \to f_*\mathcal{G}$ by $(f_*\varphi)_U := \varphi_{f^{-1}[U]}$ for each open $U \subseteq Y$. □

Example A.13. A closed affine algebraic variety X is isomorphic, as a topological space, to the maximal spectrum of its coordinate ring via the map $F(p) := \ker(\mathrm{ev}_p)$. The sheaf of regular functions $\mathcal{O}_X$ on X thus defines a sheaf $F_*\mathcal{O}_X$ on $Y := \mathrm{Spec}_{\mathfrak{m}}(\mathcal{O}(X))$ via the pushforward functor F_*. Since the basic open set $\mathfrak{D}(f) := \{\mathfrak{p} \in Y \mid f \notin \mathfrak{p}\} \subseteq Y$ for some $f \in \mathcal{O}(X)$ corresponds to the basic open $D(f) := \{p \in X \mid f(p) \neq 0\}$ under F, we may compute that the pushforward $F_*\mathcal{O}_X$ takes values

$$(F_*\mathcal{O}_X)(\mathfrak{D}(f)) := \mathcal{O}_X(D(f)) := \mathcal{O}(X)_f.$$

So the pushforward $F_*\mathcal{O}_X$ agrees with the structure sheaf $\mathcal{O}_Y$ on Y. ♢

We then run into a small issue... if we have a sheaf $\mathcal{G}$ on Y, how do we get a sheaf on X? Well, what we *want* to do is to define sections at U to be $\mathcal{G}(f[U])$, but $f[U]$ may not necessarily be open! So we may then guess that we want to approximate $f[U]$ by opens of Y and define sections as $\mathrm{colim}_{V \supseteq f[U]} \mathcal{G}(V)$. While this does indeed define a presheaf on X, it is not necessarily a sheaf. But we can fix this use étalé spaces!

Let $f^{-1}_{\mathrm{psh}}\mathcal{G}$ be the presheaf defined by

$$\left(f^{-1}_{\mathrm{psh}}\mathcal{G}\right)(U) := \operatorname*{colim}_{V \supseteq f[U]} \mathcal{G}(V).$$

Definition A.14. If $f : X \to Y$ is continuous and $\mathcal{G}$ is a sheaf on Y, then the *pullback* $f^{-1}\mathcal{G}$ of $\mathcal{G}$ along f is defined as the sheafification $f^{-1}\mathcal{G} := \left(f^{-1}_{\mathrm{psh}}\mathcal{G}\right)^+$ of the presheaf $f^{-1}_{\mathrm{psh}}\mathcal{G}$.

Though the construction of the pullback sheaf is much more involved than that of the pushforward, it still has some nice properties, such as:

Proposition A.15. Let $f : X \to Y$ be continuous and $\mathcal{G}$ a sheaf on Y. Then the stalk of the pullback $f^{-1}\mathcal{G}$ at $x \in X$ is given by

$$\left(f^{-1}\mathcal{G}\right)_x = \mathcal{G}_{f(x)}.$$

Proof. Since the sheafification of a presheaf preserves stalks (by construction), we need only check that the stalks of $f^*_{\text{psh}}\mathcal{G}$ are of this form. At $x \in X$, we see that

$$\left(f^{-1}\mathcal{G}\right)_x = \left(f^{-1}_{\text{psh}}\mathcal{G}\right)_x := \operatorname*{colim}_{U \ni x} \operatorname*{colim}_{V \supseteq f[U]} \mathcal{G}(V) = \operatorname*{colim}_{V \ni f(x)} \mathcal{G}(V) =: \mathcal{G}_{f(x)}. \qquad \square$$

Theorem A.16. *Let $f : X \to Y$ be a continuous map, then we have an adjunction $f^{-1} \dashv f_*$ of functors between $\mathbf{Sh}(X)$ and $\mathbf{Sh}(Y)$. That is, we have canonical isomorphisms*

$$\mathrm{Hom}_{\mathbf{Sh}(X)}(f^{-1}\mathcal{G}, \mathcal{F}) \cong \mathrm{Hom}_{\mathbf{Sh}(Y)}(\mathcal{G}, f_*\mathcal{F})$$

which are natural in both $\mathcal{G} \in \mathbf{Sh}(Y)$ and $\mathcal{F} \in \mathbf{Sh}(X)$.

A.3 Exactness for sheaves

Since the 'image' of a sheaf is not necssarily a sheaf, it is no longer immediately obvious how to define mono-/epi-/isomorphisms of sheaves. We may first guess that a morphism of sheaves φ is mono/epi/iso if and only if all of its components φ_U is mono/epi/iso, but this turns out not to be the useful notion.

Definition A.17. A morphism $\varphi : \mathcal{F} \to \mathcal{G}$ of sheaves of on X is said to be a *monomorphism*/*epimorphism*/*isomorphism* if the induced maps $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ on stalks are injective/surjective/bijective, respectively, for all $x \in X$. Similarly, a sequence

$$\cdots \longrightarrow \mathcal{F}_{i-1} \xrightarrow{\varphi_{i-1}} \mathcal{F}_i \xrightarrow{\varphi_i} \mathcal{F}_{i+1} \longrightarrow \cdots$$

of sheaves is said to be *exact* if the corresponding sequences of stalks are exact for all $x \in X$.

As it turns out, injectivity of induced maps of stalks is equivalent to injectivity of local components φ_U at all opens $U \subseteq X$, so the notion of a subsheaf is exactly what we think it is: $\mathcal{F}$ is a *subsheaf* of $\mathcal{G}$ if $\mathcal{F}(U)$ is a subset (subgroup, subring, etc.) of $\mathcal{G}(U)$ for each U and the restriction maps agree.

Definition A.18. If $\mathcal{F}, \mathcal{G} \in \mathbf{Sh}(X)$ are sheaves of abelian groups (or modules, rings, etc.) then the *kernel* of a morphism $\varphi : \mathcal{F} \to \mathcal{G}$ is the subsheaf $\mathcal{K}\mathrm{er}(\varphi)$ of $\mathcal{F}$ whose sections $\mathcal{K}\mathrm{er}(\varphi)(U)$ are exactly the kernel $\ker(\varphi_U)$ of the component φ_U of φ at $U \subseteq X$.

This satisfies some of the basic properties you think of, such as...

Proposition A.19. Suppose that $\varphi : \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves of abelian group (or modules, rings, etc.) on X. The following are equivalent:

(i) φ is injective;

(ii) $\mathcal{K}\mathrm{er}(\varphi)$ is the zero sheaf;

(iii) the sequence $0 \to \mathcal{F} \xrightarrow{\varphi} \mathcal{G}$ is exact at $\mathcal{F}$.

Similarly, φ is surjective if and only if $\mathcal{F} \xrightarrow{\varphi} \mathcal{G} \to 0$ is exact at $\mathcal{G}$.

A.4 Ringed spaces

We will be especially interested in rings of commutative rings. When a space has a sheaf of (commutative) rings, we call it a *ringed space*, and when its stalks are local we say that X is *locally ringed*.

Definition A.20. A *ringed space* is a pair $(X, \mathcal{O}_X)$, where X is a topological space, and a sheaf $\mathcal{O}_X$ of commutative rings on X, called its *structure sheaf*. If all of the stalks $\mathcal{O}_{X,x}$ of $\mathcal{O}_X$ are local rings (i.e. have a unique maximal ideal), then we say that $(X, \mathcal{O}_X)$ is a *locally ringed space*.

We sometimes omit $\mathcal{O}_X$ from the notation and just say that X is a (locally) ringed space.

Example A.21. (i) Let X be a space and define its structure sheaf $\mathcal{O}_X$ to be the sheaf of continuous functions to $\mathbb{R}$. Then since $\mathbb{R}$ is a commutative topological ring, each $\mathcal{O}_X(U)$ inherits a commutative ring structure. The stalk $\mathcal{O}_{X,x}$ at $x \in X$ has its ring structure explicitly given by

$$[s,U] + [t,V] = [s|_{U\cap V} + t|_{U\cap V}, U \cap V], \qquad [s,U] \cdot [t,V] = [(s|_{U\cap V}) \cdot (t|_{U\cap V}), U \cap V],$$

and a germ $[s,U]$ is invertible if and only if and only if $s(x) \neq 0$, by continuity. Let $\mathfrak{m}_x$ denote the set of germs $[s,U]$ at x for which $s(x) = 0$, which is seen to be an ideal of $\mathcal{O}_{X,x}$. Clearly this is maximal, since anything not in $\mathfrak{m}_x$ is a unit. Moreover, if J is a proper ideal of $\mathcal{O}_{X,x}$ then it contains no units, so is contained in $\mathfrak{m}_x$. Consequently, $\mathfrak{m}_x$ is the unique maximal ideal of $\mathcal{O}_{X,x}$, and so $\mathcal{O}_{X,x}$ is local. So $(X, \mathcal{O}_X)$ is a locally ringed space;

(ii) If M is a smooth manifold, then the sheaf $\mathcal{O}_M$ of smooth functions to $\mathbb{R}$ is also seen make M a locally ringed space, for exactly the same reasons as above;

(iii) If V is an algebraic variety (or scheme), then the sheaf $\mathcal{O}_V$ of regular functions (resp. morphisms) to $\mathbb{A}^1$ makes V a locally ringed space for the same reasons.

♢

Definition A.22. A *morphism* $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ *of ringed spaces* consists of a continuous map $f : X \to Y$ and a sheaf morphism $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$. If, additionally, X and Y are *locally* ringed spaces, then we say that $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a *morphism of locally ringed spaces* if the induced maps $f^\#_p : \mathcal{O}_{Y,p} \to (f_*\mathcal{O}_X)_p$ of stalks are all local homomorphisms, i.e. the maximal of $\mathcal{O}_{Y,p}$ is mapped into the maximal of $(f_*\mathcal{O}_X)_p$.

A.4.1 $\mathcal{O}_X$-modules on ringed spaces

If we put another sheaf on a ringed space, we may wish that it is compatible with the structure sheaf $\mathcal{O}_X$ in some way. This naturally leads to the notion of an *$\mathcal{O}_X$-module*.

Definition A.23. Let $(X, \mathcal{O}_X)$ be a ringed space. An *$\mathcal{O}_X$-module* is a sheaf $\mathcal{F}$ of abelian groups on X, along with a compatible $\mathcal{O}_X(U)$-module structure on $\mathcal{F}(U)$ for each open $U \subseteq X$. Here 'compatible' just means means that if $U \supseteq V$ are opens then the following diagram commutes:

$$\begin{array}{ccc} \mathcal{O}_X(U) \times \mathcal{F}(U) & \longrightarrow & \mathcal{F}(U) \\ {\scriptstyle \rho^U_V \times \rho^U_V}\downarrow & & \downarrow{\scriptstyle \rho^U_V} \\ \mathcal{O}_X(V) \times \mathcal{F}(V) & \longrightarrow & \mathcal{F}(V) \end{array}$$

where the horizontal arrows are those arising from the $\mathcal{O}_X(U)$-module structure on $\mathcal{F}(U)$, and the same for V. An *$\mathcal{O}_X$-algebra* is a sheaf $\mathcal{F}$ of rings which is also an $\mathcal{O}_X$-module.

One checks that this diagram commuting just says that the restriction $\rho^U_V(\lambda m)$ of the product λm is equal to the product $\rho^U_V(\lambda)\rho^U_V(m)$ of the respective restrictions, where $\lambda \in \mathcal{O}_X(U)$ and $m \in \mathcal{F}(U)$.

Example A.24. (i) Let X be a space, R a commutative ring, and M a module over R. We may equip X with the constant sheaf $\underline{R}$, making $(X, \underline{R})$ a ringed space (though not necessarily locally ringed). Then the constant sheaf $\underline{M}$ on $(X, \underline{R})$ becomes an $\underline{R}$-module;

(ii) If X is a complex manifold, let $\mathcal{O}_X = \mathrm{Hol}(-, \mathbb{C})$ denote the sheaf of holomorphic functions. This makes $(X, \mathcal{O}_X)$ a locally ringed space. The sheaf $\mathcal{M} := \mathrm{Mer}(-, \mathbb{C})$ of meromorphic functions is then an $\mathcal{O}_X$-algebra.

♢

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